Artificial Intelligence and Big Data

Postgraduate Program in Central Banking (CEMFI)

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About this Course

This course provides a hands-on introduction to artificial intelligence and big data. Although we will explore fundamental theoretical concepts to better comprehend the techniques employed, the primary focus is on applied implementation. Python will be our programming language of choice throughout the course, as it is the leading language in the field.

This course is divided into four parts, each focusing on different aspects of artificial intelligence and big data. The expected time for each part includes both theoretical and practical components, with a total of approximately 24.5 hours.

- **Part I: Foundations**
  1. Introduction to Artificial Intelligence and Big Data
  2. Introduction to Python
- **Part II: Supervised Machine Learning**
  3. Overview of Supervised Learning
  4. Decision Trees
  5. Neural Networks
  6. Practice Session I
- **Part III: Natural Language Processing**
  7. Overview of Natural Language Processing (NLP)
  8. Classical NLP Approaches
  9. Practice session II
- **Part IV: Generative AI**
  10. Overview of Generative AI
  11. Large Language Models (LLMs)
  12. Practice session III

Useful Resources

The course does not follow a particular textbook but has drawn material from several sources such as

- Hastie, Tibshirani, and Friedman (2009), "The Elements of Statistical

Learning"
- Murphy (2012), "Machine Learning: A Probabilistic Perspective"
- Murphy (2022), "Probabilistic Machine Learning: An Introduction"
- Murphy (2023), "Probabilistic Machine Learning: Advanced Topics"
- Goodfellow, Bengio, and Courville (2016), "Deep Learning"
- Bishop (2006), "Pattern Recognition And Machine Learning"
- Nielsen (2019), "Neural Networks and Deep Learning"
- Sutton and Barto (2018), "Reinforcement Learning: An Introduction"

Note that all of these books are officially **available for free** in the form of PDFs or online versions (see the links in the references). However, you are not required to read them and, as a word of warning, the books go much deeper into the mathematical theory behind the machine learning techniques than we will in this course. Nevertheless, you may find them useful if you want to learn more about the subject.

Regarding **programming in Python**, McKinney (2022) "Python for Data Analysis" might serve as a good reference book. The book is **available for free** online and covers a lot of the material we will be using in this course. You can find it here: Python for Data Analysis.

Software Installation Notes

We will be using Python for this course. For simplicity, we will be using the Anaconda distribution, which is a popular distribution of Python (and R) that aims to simplify the management of packages. We will also be using the Visual Studio Code (VS Code) as our code editor.

Anaconda Installation

The first step is to install the Anaconda distribution:

1. Download the Anaconda distribution from anaconda.com. Note: If you are using a M1 Mac (or newer), you have to choose the 64-Bit (M1) Graphical Installer. With an older Intel Mac, you can choose the 64-Bit Graphical Installer. With Windows, you can choose the 64-Bit Graphical Installer (i.e., the only Windows option).

2. Open the installer that you have downloaded in the previous step and follow the on-screen instructions.

3. If it asks you to update Anaconda Navigator at the end, you can click `Yes` (to agree to the update), `Yes` (to quit Anaconda Navigator) and then `Update Now` (to actually start the update).

To **confirm that the installation was successful**, you can open a *terminal window* on macOS/Linux or an *Anaconda Prompt* if you are on Windows and

run the following command:

```
conda --version
```

This should display the version of Conda that you have installed. If you see an error message, the installation was likely not successful and you should ask for advice from your peers or send me an email.

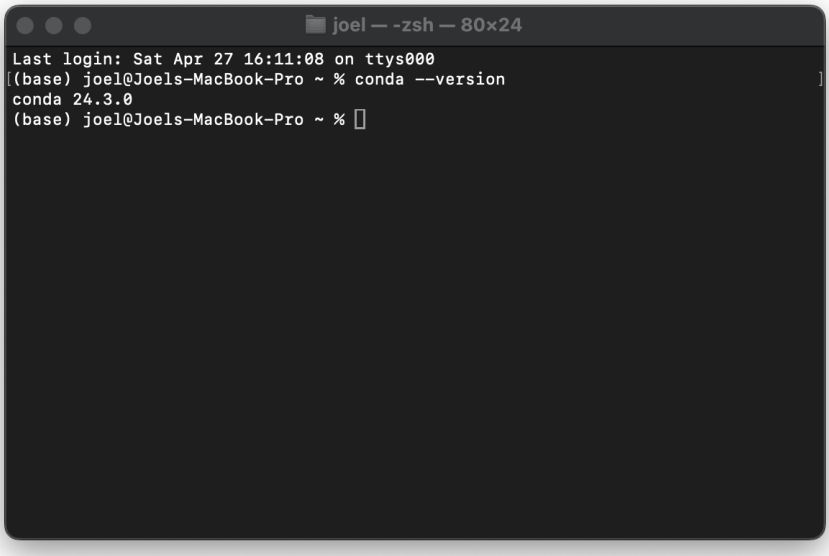


Figure 1: Terminal Output after Anaconda Installation

Creating a Conda Environment

Next, we want to create a new environment for this course that contains the correct Python version and all the Python packages we need. We can do this by creating a new Conda environment from the `environment.yml` provided on Moodle.

1. Open a *terminal window* on macOS/Linux or an *Anaconda Prompt* if you are on Windows.

2. There are two ways to create the Conda environment:

   Option A: Run the following command from the terminal or Anaconda Prompt:

   ```
   conda env create -f https://datascience.joelmarbet.com/environment.yml
   ```

This downloads the `environment.yml` file automatically and creates the environment.

Option B: Download the `environment.yml` file manually:

i. Navigate to the folder where you have downloaded the `environment.yml` file. On macOS/Linux, you can do this by running the following command in the terminal:

```
cd ~/Downloads
```

which will navigate to the `Downloads` folder in your home directory.

On Windows, you can do this by running the following command in the Anaconda Prompt:

```
cd "%userprofile%/Downloads"
```

which will navigate to the `Downloads` folder in your user profile.

Note that if you use a different path that contains space you need to put the path in quotes, e.g., `cd "~/My Downloads"`.

ii. Create a new Conda environment from the `environment.yml` file by running the following command in the terminal or Anaconda Prompt:

```
conda env create -f environment.yml
```

Either option will create a new Conda environment called `datascience_course_cemfi` with the correct Python version and all the Python packages we need for this course. Note that the installation might take a few minutes.

3. Activate the new Conda environment by running the following command in the terminal or Anaconda Prompt:

```
conda activate datascience_course_cemfi
```

To **confirm that the environment was created successfully**, you can run the following command in the terminal or Anaconda Prompt:

```
python --version
```

This should display Python version 3.8.8. If you see another Python version you might have forgotten to activate the environment or the environment was not created successfully.

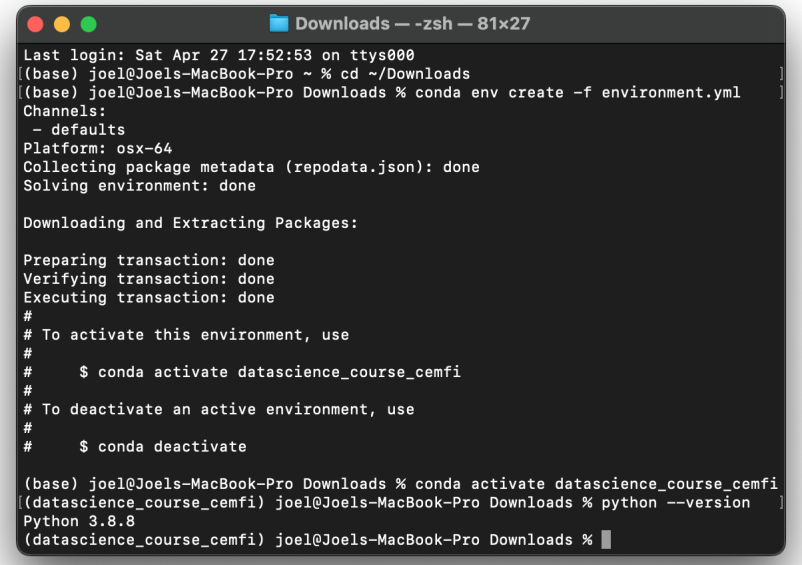


Figure 2: Terminal Output From Environment Creation

Resetting or Updating a Conda Environment

If you accidentally make changes to the environment and want to reset it to the original state, you can do this by navigating to the folder where you have downloaded `environment.yml` and then running the following command in the *terminal* or *Anaconda Prompt*:
`conda env update --file environment.yml --prune`
Alternatively, you can also update the environment by running the following command in the *terminal* or *Anaconda Prompt*, which downloads the environment.yml file automatically from the course website:
`conda env update --file https://datascience.joelmarbet.com/environment.yml --prune`
This can also be used to update the environment if we add new packages to the `environment.yml` file.

Installing VS Code

The last step is to install the Visual Studio Code (VS Code) editor:

1. Download the Visual Studio Code editor from code.visualstudio.com.
2. Open the installer that you have downloaded in the previous step and follow the on-screen instructions.

We also need to install some VS Code extensions that will help us with Python programming and Jupyter notebooks:

1. Open VS Code.

2. Click on the `Extensions` icon on the left sidebar (or press `Cmd+Shift+X` on macOS or `Ctrl+Shift+X` on Windows).

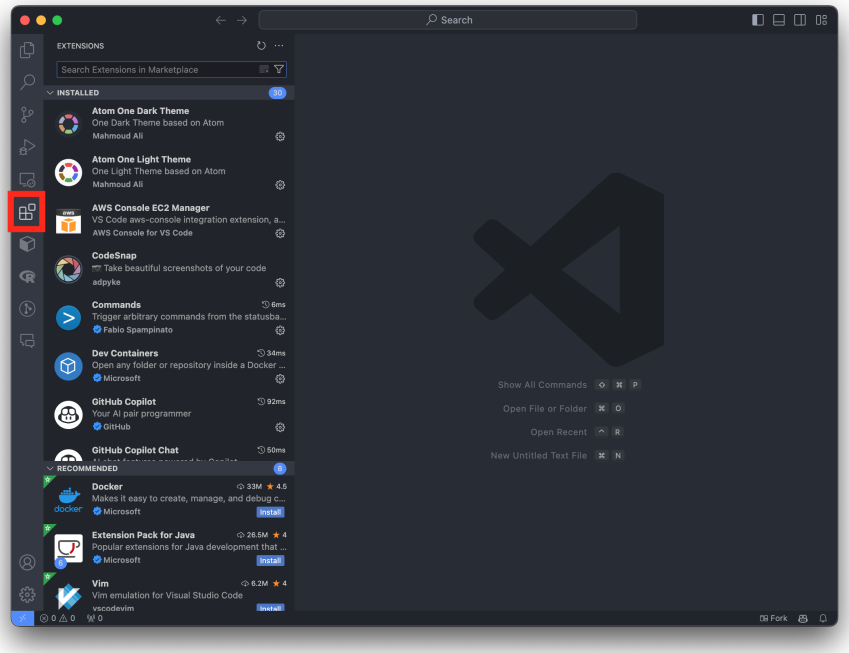


Figure 3: Installing Extensions in VSCode

3. Search for `Python` and click on the `Install` button for the extension that is provided by Microsoft.

4. Search for `Jupyter` and click on the `Install` button for the extension that is provided by Microsoft.

Testing the Installation

To test the installation, you can download a Juypter notebook from Moodle and open it in VS Code:

1. Open the Jupyter notebook in VS Code.
2. Click on `Select Kernel` in the top right corner of the notebook and choose the `datascience_course_cemfi` kernel.

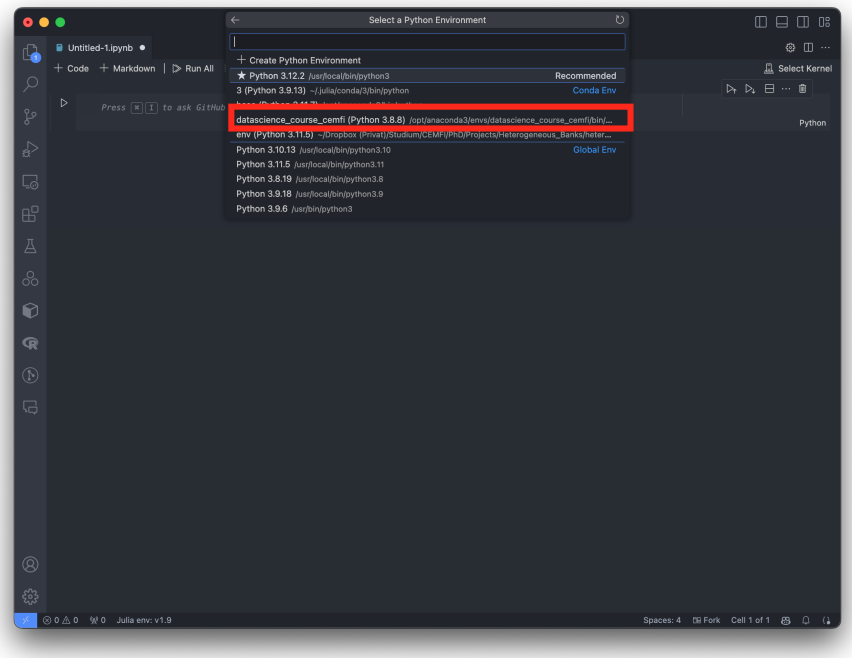


Figure 4: VSCode Jupyter Kernel Selection

3. Run the first cell of the notebook by clicking on the `Execute Cell` button next to the cell on the left.

If you see the output of the cell (or a green check mark below the cell), the installation was successful.

> **Running Jupyter Notebooks in the Browser**
>
> If you have issues running Jupyter notebooks in VSCode, you can also run them in the browser. To do this, you can open a *terminal window* on macOS/Linux or an *Anaconda Prompt* if you are on Windows and run the

following command:
`jupyter notebook`
This will open a new tab in your default browser with the Jupyter notebook interface. You can then navigate to the folder where you have downloaded the course materials and open the notebooks from there.

Part I

Foundations

Chapter 1

Introduction

This is a book created from markdown and executable code.

See (**?**) for additional discussion of literate programming.

Chapter 2

Programming in Python

This section provides a brief introduction to programming in Python, covering the basics of the language, essential libraries for data analysis, and best practices for coding. The goal is to equip you with the skills needed to work with Python effectively in the context of artificial intelligence and big data.

Python has become the de facto standard for AI and data science due to its simplicity, readability, and rich ecosystem of specialized libraries. Throughout the course, we will use Python for various tasks, including data manipulation, visualization, statistical analysis, and implementing machine learning algorithms. By the end of this section, you should be comfortable with Python's core concepts and ready to tackle basic real-world AI and big data challenges.

Note that programming is a skill that cannot be mastered overnight. It requires practice and continuous learning. I encourage you to experiment with the code examples provided in this section and to work through the exercises. Don't worry if things don't click immediately; programming fluency develops through repetition and problem-solving.

> **Note**
>
> The material in this section draws from the material developed by Alba Miñano and extended by Jesús Villota Miranda, which they kindly prepared for another data science course that I taught at CEMFI.

2.1 Overview of Python

2.1.1 What is Python?

Python is a high-level, interpreted programming language created by Guido van Rossum and first released in 1991. It emphasizes code readability and simplicity through its clean syntax and use of significant whitespace, making it an ideal language for both beginners and experienced programmers. Python is a general-purpose language that excels across diverse domains—from automation to scientific computing and artificial intelligence. Its extensive standard library and vast ecosystem of third-party packages enable rapid development and prototyping. Today, Python is one of the most popular programming languages worldwide and has become the lingua franca of data science and machine learning, largely due to powerful libraries like NumPy, Pandas, scikit-learn, TensorFlow, and PyTorch.

2.1.2 Why Python and Not Other Languages?

While languages like R, Julia, and MATLAB, or sometimes even lower level languages like C++ are used in data science and AI, Python offers distinct advantages for this course:

- **Unified ecosystem**: Python handles the entire data science workflow—from data collection and cleaning to modeling and deployment—within a single language, avoiding the friction of switching between tools.
- **Industry adoption**: Major tech companies and research institutions have standardized on Python for AI/ML work, making it the most marketable skill for practitioners.
- **Library maturity**: The ecosystem offers battle-tested libraries (NumPy, Pandas, scikit-learn) alongside cutting-edge deep learning frameworks (PyTorch, TensorFlow), providing both stability and innovation.
- **Gentle learning curve**: Readable syntax allows you to focus on concepts rather than wrestling with language complexity—particularly valuable when learning AI and big data techniques.
- **Community support**: With the largest community in data science, you'll find abundant tutorials, Stack Overflow answers, and open-source projects to learn from.

That said, other languages have their strengths: R, for example, excels in statistical analysis and visualization and Julia offers superior performance for numerical computing. Python strikes the best balance for our purposes: **accessible enough for newcomers yet powerful enough for production systems**.

2.1.3 Installation and Setup

For this course, we will primarily use Nuvolos, a cloud-based platform that provides a pre-configured Python environment with all necessary libraries and a VS Code interface. This eliminates installation headaches and ensures everyone has an identical setup. You can access Nuvolos through the link in the sidebar.

However, learning to set up Python locally is a valuable skill for future projects. If you wish to work on your own machine, here are the general steps to install Python and the required packages:

1. **Install Python via Anaconda/Miniconda**: Anaconda is a distribution that bundles Python with common data science packages. Miniconda is a lighter version that installs only Python and the conda package manager, allowing you to install packages as needed.
2. **Create a virtual environment**: Virtual environments isolate project dependencies, preventing version conflicts between projects. Use `conda env create -f https://ai-bigdata.joelmarbet.com/environment.yml` to create the environment required for this course.
3. **Install additional packages**: If required, you can install additional packages individually (e.g., `conda install numpy`).
4. **Set up VSCode**: Install Visual Studio Code and add the Python and Jupyter extensions. VSCode provides an excellent development experience with features like code completion, debugging, and integrated notebook support.

Detailed installation instructions on how to install the environment used in this course are available in the "Notes for Local Installation" PDF linked in the sidebar. For troubleshooting or platform-specific issues, consult the documentation or reach out after class or by email.

2.2 Development Environment

A good development environment significantly improves your productivity and learning experience. This section covers the main tools you'll encounter in this course.

2.2.1 Visual Studio Code (VSCode)

VSCode is a free, lightweight, yet powerful code editor that has become a developer favorite for many different programming languages. It combines the simplicity of a text editor with features traditionally found in full IDEs.

Figure 2.1 shows the main components of the VSCode interface:

1. **Activity Bar**: Located on the far left, it provides quick access to different views like Explorer, Search, Source Control, Extensions, and more.

2. **Side Bar**: Displays different views depending on the selected activity (e.g., file explorer, search results).
3. **Editor Area**: The central area where you write and edit your code files.
4. **Panel**: The bottom area that can display output, terminal, problems, and debug information.
5. **Status Bar**: Located at the bottom, it shows information about the current file, such as line number, encoding, and selected Python interpreter.

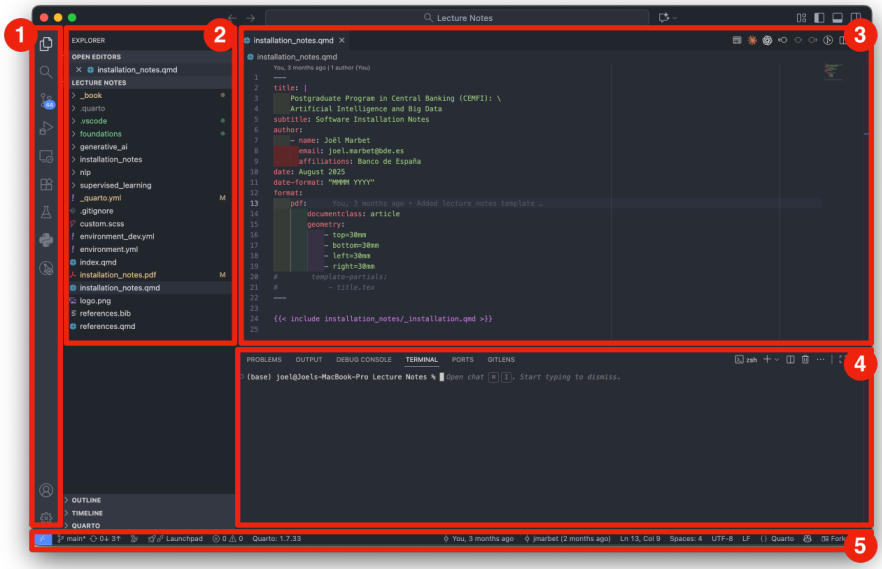


Figure 2.1: VSCode - Overview

Note that not all elements are always visible; for example, the Panel is hidden by default and can be toggled as needed.

For Python development, you'll want to install the following extensions in VSCode:

- **Python Extension**: Provides IntelliSense (code completion), linting, debugging, and code navigation. It automatically detects your Python installations and allows you to select interpreters.
- **Jupyter Extension**: Enables you to create, edit, and run Jupyter notebooks directly within VSCode, eliminating the need to switch to a browser.

You can install extensions by clicking on the Extensions icon in the left sidebar and searching for them by name.

2.3 Install Python Extension

Figure 2.2: VSCode - Install Python Extension

2.4 Install Jupyter Extension

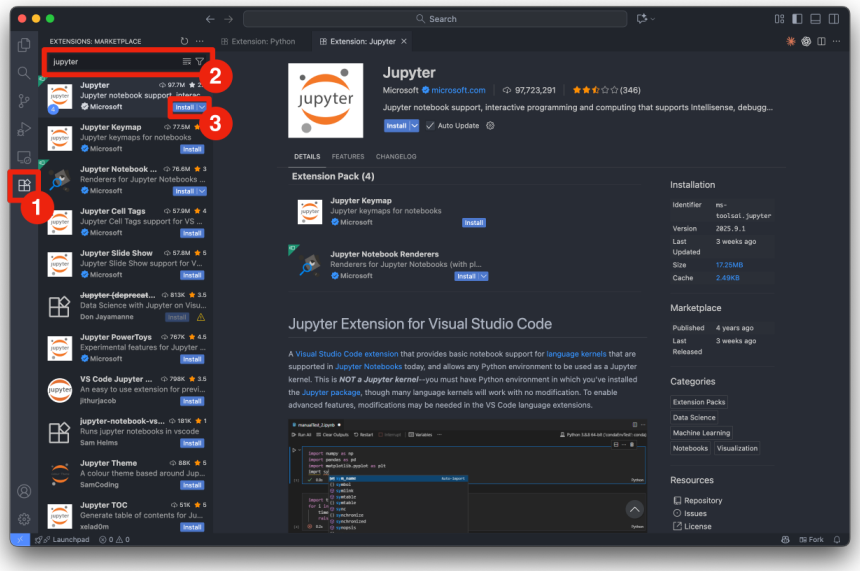


Figure 2.3: VSCode - Install Jupyter Extension

We will primarily use VSCode within Nuvolos for this course, but you can also set it up locally following the installation instructions provided earlier. Note that the version on Nuvolos has an additional menu button at the top left, which provides access to menues to open files, settings, and other options. In the local version of VSCode, these options are available in the standard menu bar at the top of the window/screen.

2.4.1 Jupyter Notebooks

The main way we will interact with Python code in this course is through Jupyter Notebooks. Jupyter Notebooks, as well as the popular Jupyter Lab, are all part of the **Jupyter Project** which revolves around the provision of tools and standards for interactive computing across different computing languages (**Ju**lia,**Py**thon,**R**).

Jupyter Notebooks are interactive documents that combine live code, visualizations, and explanatory text. They're ideal for exploratory data analysis and prototyping. They allow you to write and execute code in small chunks (cells), see immediate outputs, and document your thought process alongside the code. While Jupyter Notebooks are excellent for exploration and learning, they may not be the best choice for production code or large projects due to challenges

with version control and code organization. However, they remain a popular tool in data science and AI for their interactivity and ease of use. We will execute Jupyter Notebooks within VSCode instead of the more traditional browser-based interface. The reason for this choice is to provide a unified development environment where you can seamlessly switch between writing notebooks and scripts, debugging code, and managing files. Furthermore, VSCode integrates well with recent AI-assisted coding tools, which can enhance your productivity.

Figure 2.4 shows an example of a Jupyter notebook opened in VSCode. To work with Jupyter notebooks in VSCode, follow these steps:

1. **Open Notebook**: Open a notebook file (file extension: `.ipynb`) or create a new one from the menu ("File" -> "New File" and then select "Jupyter Notebook").
2. **Choose Kernel**: Ensure that the kernel (Python interpreter) is set correctly. In this course, you should always select `ai-big-data-cemfi` as the kernel. You can change the kernel by clicking on the current kernel name (or "Select Kernel") in the top-right corner of the notebook interface (denoted by number 1 in Figure 2.4). Then, click on "Python Environments" and select `ai-big-data-cemfi` from the list.

If you have done this correctly, you should see `ai-big-data-cemfi` displayed as the selected kernel as shown in Figure 2.4.

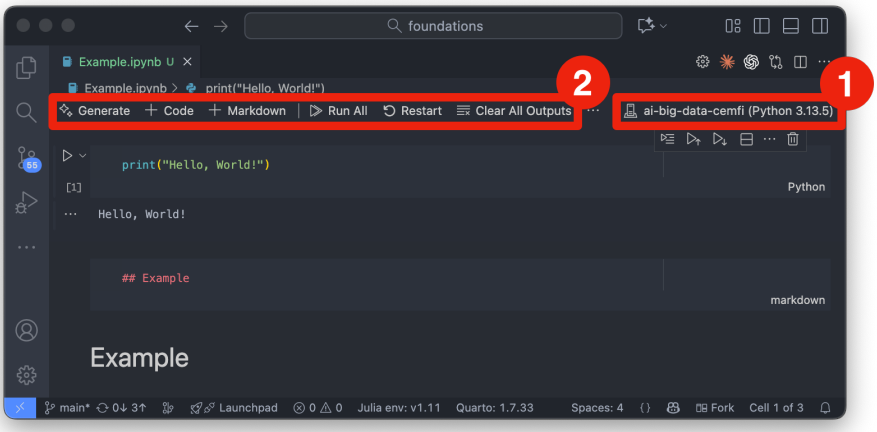


Figure 2.4: VSCode - Jupyter Initial Setup

A Jupyter notebook consists of a sequence of cells, which can be of two main types:

- **Code Cells**: These cells contain executable code. You can run them individually, and the output (results, plots, error messages) will be displayed directly below the cell.

- **Markdown Cells**: These cells allow you to write formatted text using Markdown syntax. You can include headings, lists, links, images, and even LaTeX equations for mathematical notation.

These cells can be created from the toolbar at the top of the notebook interface (denoted by number 2 in Figure 2.4) or from the + button appearing under cells when hovering over them. From the toolbar you can also run cells, stop execution, restart the kernel, and perform other notebook-related actions. Cells can also be executed by selecting them and pressing `Shift-Enter` or by clicking the "Play" button in the toolbar (denoted by number 4 in Figure 2.5). Once you run a cell, the output will appear directly below it (denoted by number 5 in Figure 2.5). Markdown cells can be edited by double-clicking on them, and you can switch between code and markdown cell types using the dropdown menu in the toolbar. Number 2 and 3 in Figure 2.5 show markdown cells that are being edited and rendered, respectively. Number 1 in Figure 2.5 shows a code cell. Note that code cells have "Python" written in the bottom right corner to indicate the language being used.

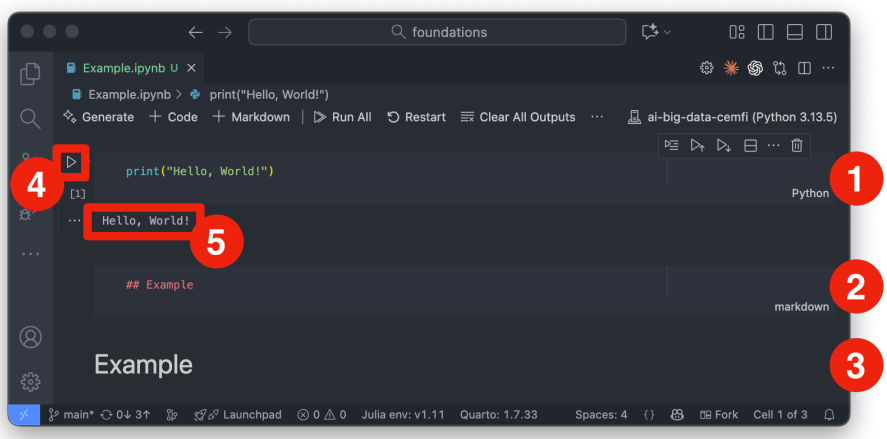


Figure 2.5: VSCode - Jupyter Cells

2.4.2 Notebooks vs. Scripts

Another way to write and run Python code is through scripts—plain text files with a `.py` extension that contain Python code. Scripts are executed as a whole, either from the command line or within an IDE like VSCode. They are better suited for larger projects, production code, and automation tasks.

Figure 2.6 shows an example of a Python script opened in VSCode. You can run the entire script by right-clicking anywhere in the editor and selecting "Run Python File in Terminal" or by clicking the "Play" button (denoted by number 1

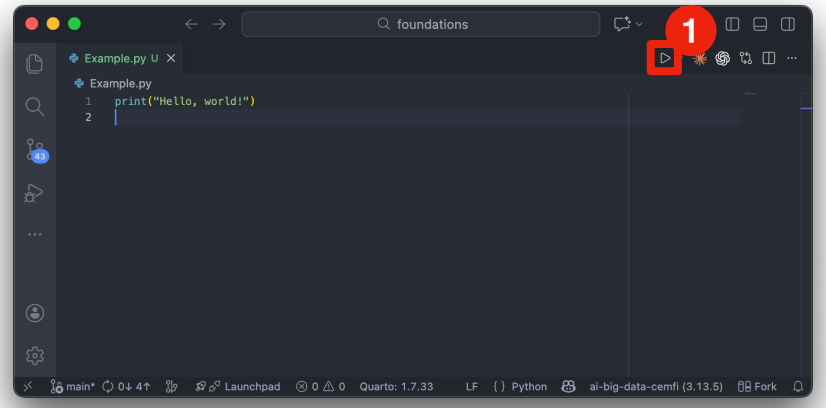


Figure 2.6: VSCode - Python Script

in Figure 2.6). The output will appear in the integrated terminal at the bottom of the VSCode window.

When to use notebooks:

- Exploratory data analysis and visualization
- Step-by-step tutorials and documentation
- Quick prototyping and experimentation
- Presenting results with integrated plots and explanations

When to use scripts:

- Production code and automation
- Code that will be imported as modules
- Version control such as Git (notebooks can be harder to diff)
- Long-running processes without intermediate outputs

Best Practices:

- Keep notebooks focused on a single topic or analysis
- Use descriptive cell outputs and markdown for documentation
- Restart kernel and run all cells before sharing to ensure reproducibility

We will primarily use Jupyter notebooks for in-class exercises and exploratory tasks, but I will provide some Python scripts as examples. Understanding both formats is important for effective Python programming.

2.4.3 Google Colab

Google Colab is a free cloud-based Jupyter notebook environment that requires no setup and provides free access to GPUs. It's particularly useful for:

- Working on machines without Python installed
- Experimenting with deep learning models that require GPUs
- Collaborating with others in real-time (similar to Google Docs)
- Accessing more computational resources than your local machine provides

Limitations:

- Sessions timeout after periods of inactivity
- Files are stored in Google Drive or must be re-uploaded each session
- Not suitable for long-running jobs or production workflows
- Might not work in certain restricted corporate environments

If you have trouble installing the environments locally, Google Colab can be a good alternative. To use Colab, simply navigate to colab.research.google.com. There you can create a new notebook or upload an existing one. I will provide links to the notebooks used in this course that you can open directly in Colab if needed. However, Nuvolos is the preferred environment for this course and will give you the best experience.

2.5 Python Fundamentals

Python is an **interpreted language**. By this we mean that the Python interpreter will run a program by executing the source code line-by-line without the need for compilation into machine code beforehand. Furthermore, Python is an **Object-Oriented Programming (OOP)** language. Everything we define in our code exists within the interpreter as Python object, meaning it will have its associated **attributes** and **methods**. We will see these concepts in more detail later.

First, let's have a look at the basics of any programming language. All programs consist of the following

- Variables,
- Functions,
- Loops, and
- Conditionals.

2.5.1 Variables

Variables are basic elements of any programming language. They

- store information,
- can be manipulated by the program, and

- can be of different types, e.g. integers, floating point numbers (floats), strings (sequences of characters), or booleans (true or false)

2.5.1.1 Creating Variables

Python is **dynamically typed**, meaning you don't need to declare variable types explicitly. The interpreter infers the type based on the assigned value. For example, the following code creates a variable `x` and assigns it the integer value `100`. The `type()` function is then used to check the type of the variable.

```
x = 100
type(x)
```

```
int
```

The python interpreter output `int`, indicating that `x` is of type integer.

As the example above shows, you can create a variable by simply assigning a value to it using the equals sign (`=`). What happens under the hood is that Python creates an object in memory to store the value `100` and then creates a reference (the variable name `x`) that points to that object. When you later use the variable `x` in your code, Python retrieves the value from the memory location that `x` references. For example, we can then do computations with `x`:

```
y = x + 50
print(y)
```

```
150
```

Python retrieved the value of `x` (which is `100`), added `50` to it, and assigns the result to the new variable `y`.

Note that you can reassign variables to new values or even different types. For example, you can change the value of `x` simply by assigning a new value to it

```
x = 200
print(x)
```

```
200
```

Note that now `x` points to a new object in memory with the value `200`. The previous object with the value `100` will be automatically cleaned up by Python's garbage collector if there are no other references to it. This might not seem important now, but there are some implications of this behavior when working with mutable objects, which we will cover later.

2.5.1.2 Naming Variables

The process of naming variables is an important aspect of programming. Good variable names enhance code readability and maintainability, making it easier

for others (and yourself) to understand the purpose of each variable.

For example, consider the following two variable names

```
a = 25
number_of_students = 25
```

The first variable name, `a`, is vague and does not convey any information about what it represents. In contrast, `number_of_students` is descriptive and clearly indicates that the variable holds the count of students. This makes the code more understandable, especially in larger programs where many variables are used.

Python imposes certain rules on how variable names can be constructed

1. They must start with a letter (a-z, A-Z) or an underscore (_).
2. They can only contain letters, numbers (0-9), and underscores.
3. They cannot be the same as Python's reserved keywords (e.g., `if`, `else`, `while`, `for`, etc.). `help(keywords)` will show which words are reserved.
4. Variable names are case-sensitive, meaning that `Variable`, `variable`, and `VARIABLE` would be considered different variables.

In addtition to these rules, good practices for naming variables include to

- Use meaningful and descriptive names that convey the purpose of the variable
- Use lowercase letters and separate words with underscores (`snake_case`) for better readability (some programmers use `camelCase`, but snake_case is preferred in Python)
- Avoid using single-letter names except for loop counters or very short-lived variables
- Avoid using built-in function names because that will overwrite the function (i.e., if we write `type` we will no longer be able to use `type` to access the 'type' of variables)
- Be consistent with naming conventions throughout your codebase
- While you can use names in any language, English is generally preferred so that anyone can follow the code

The following code snippet lists all reserved keywords in Python that cannot be used as variable names

```
import keyword

for kw in keyword.kwlist:
    print(kw)
```

```
False
None
True
and
```

```
as
assert
async
await
break
class
continue
def
del
elif
else
except
finally
for
from
global
if
import
in
is
lambda
nonlocal
not
or
pass
raise
return
try
while
with
yield
```

Make sure you don't use any of these words as variable names in your code.

2.5.1.3 Basic Data Types

Python has several built-in data types that are commonly used

- **Integers (`int`)**: Whole numbers, e.g., `42`, `-7`
- **Floating-point numbers (`float`)**: Numbers with decimal points, e.g., `3.14`, `-0.001`
- **Complex numbers (`complex`)**: Numbers with real and imaginary parts, e.g., `2 + 3j`
- **Strings (`str`)**: Sequences of characters enclosed in single or double quotes, e.g., `'Hello, World!'`, `"Python"`
- **Booleans (`bool`)**: Logical values representing `True` or `False`

Since Python is dynamically typed, the creation of variables of these types is straightforward, as shown in the following examples

```
this_is_int = 5
type(this_is_int)
```

```
int
```

```
this_is_float = 3.14
type(this_is_float)
```

```
float
```

```
this_is_complex = 2 + 3j
type(this_is_complex)
```

```
complex
```

```
this_is_str = "Hello, Python!"
type(this_is_str)
```

```
str
```

```
this_is_bool = True
type(this_is_bool)
```

```
bool
```

Note that boolean values are special in the sense that they are equivalent to integers: `True` is equivalent to `1` and `False` is equivalent to `0`. This means you can perform arithmetic operations with boolean values, and they will behave like integers in those contexts.

There is another data type called `NoneType`, which you might encounter. It represents the absence of a value and is created using the `None` keyword.

```
this_is_none = None
type(this_is_none)
```

```
NoneType
```

You can also create more complex data types, which we will cover in the section on data structures.

2.5.1.4 Basic Operations

A key element of programming is manipulating the variables you create. Python supports various basic operations for different data types, including arithmetic operations for numbers, string operations for text, and boolean operations for logical values.

Arithmetic Operations: You can perform arithmetic operations on integers and floats using operators like `+`, `-`, `*`, `/`, `//` (floor division), `%` (modulus), and `**` (exponentiation).

```
a = 10
b = 3
```

```
sum_result = a + b # Addition
print(sum_result)
```

```
13
```

```
diff_result = a - b # Subtraction
print(diff_result)
```

```
7
```

```
prod_result = a * b # Multiplication
print(prod_result)
```

```
30
```

```
div_result = a / b # Division
print(div_result)
```

```
3.3333333333333335
```

```
floor_div_result = a // b # Floor Division
print(floor_div_result)
```

```
3
```

```
mod_result = a % b # Modulus
print(mod_result)
```

```
1
```

```
exp_result = a ** b # Exponentiation
print(exp_result)
```

```
1000
```

String Operations: Strings can be concatenated using the `+` operator and repeated using the `*` operator.

```
str1 = "Hello, "
str2 = "World!"
concat_str = str1 + str2 # Concatenation
print(concat_str)
```

```
Hello, World!
```

Sometimes, you may want to repeat a string multiple times

```
repeat_str = str1 * 3 # Repetition
print(repeat_str)
```

```
Hello, Hello, Hello,
```

Another useful operation is string interpolation, which allows you to embed variables within strings. This can be done using f-strings (formatted string literals) by prefixing the string with `f` and including expressions inside curly braces `{}`.

```
name = "Alba"
age = 30
intro_str = f"Her name is {name} and she is {age} years old."
print(intro_str)
```

```
Her name is Alba and she is 30 years old.
```

Boolean Operations: You can use logical operators like `and`, `or`, and `not` to combine or negate boolean values.

```
bool1 = True
bool2 = False
and_result = bool1 and bool2 # Logical AND
print(and_result)
```

```
False
```

```
or_result = bool1 or bool2 # Logical OR
print(or_result)
```

```
True
```

```
not_result = not bool1 # Logical NOT
print(not_result)
```

```
False
```

To compare values, you can use comparison operators like `==` (equal to), `!=` (not equal to), `<` (less than), `>` (greater than), `<=` (less than or equal to), and `>=` (greater than or equal to).

```
a = 10
b = 20
```

```
eq_result = (a == b) # Equal to
print(eq_result)
```

```
False
```

```
neq_result = (a != b) # Not equal to
print(neq_result)
```

```
True
```

```
lt_result = (a < b) # Less than
print(lt_result)
```

```
True
```

```
gt_result = (a > b) # Greater than
print(gt_result)
```

```
False
```

```
le_result = (a <= b) # Less than or equal to
print(le_result)
```

```
True
```

```
ge_result = (a >= b) # Greater than or equal to
print(ge_result)
```

```
False
```

Note that the result of comparison operations is always a boolean value (`True` or `False`). This will be useful when we discuss conditional statements later.

> **Warning**
>
> Be careful not to confuse the assignment operator `=` with the equality comparison operator `==`. The single equals sign `=` assigns a value to a variable, while the double equals sign `==` checks if two values are equal and returns a boolean result.

We can also combine multiple comparison operations using logical operators. For example, to check if a number is within a certain range, we can use the `and` operator

```
num = 15
is_in_range = (num > 10) and (num < 20)
print(is_in_range)
```

```
True
```

This checks if `num` is greater than `10` and less than `20`, returning `True` if both conditions are met. Of course, we can also use `or` to check if at least one condition is met or `not` to negate a condition.

2.5.2 Functions

Functions are reusable blocks of code that perform a specific task. They help organize code, improve readability, and allow for code reuse. In Python, you define a function using the `def` keyword, followed by the function name and parentheses containing any parameters. For example, here is a simple function that takes two arguments, performs a calculation, and returns the result

```
def function_name(arg1, arg2):
  r3 = arg1 + arg2
  return r3
```

Note that the indentation (whitespace at the beginning of a line) is crucial in Python, as it defines the scope of the function. The code block inside the function must be indented consistently. In the example above, two spaces are used for indentation, but tabs or four spaces are also common conventions. VSCode will automatically convert tabs to spaces based on your settings and the convention used in the file.

Suppose we want to create a function that greets a user by their name. We can define such a function as follows

```
def greet(name):
  greeting = f"Hello, {name}!"
  return greeting
```

You can then call the function by passing the required argument

```
message = greet("Alba")
print(message)
```

```
Hello, Alba!
```

We could also define the function without a return value and simply print the greeting directly

```
def greet_print(name):
  print(f"Hello, {name}!")
```

You can call this function in the same way

```
greet_print("Alba")
```

```
Hello, Alba!
```

We can also define functions with multiple outputs by returning a tuple of values. For example, here is a function that takes two numbers and returns both their sum and product

```
def sum_and_product(x, y):
  sum_result = x + y
```

```
  product_result = x * y
  return sum_result, product_result
```

You can call this function and unpack the returned values into separate variables

```
s, p = sum_and_product(5, 10)
print(f"Sum: {s}, Product: {p}")
```

```
Sum: 15, Product: 50
```

or you can capture the returned tuple in a single variable

```
result = sum_and_product(5, 10)
print(f"Result: {result}")
```

```
Result: (15, 50)
```

You can define functions with mulitple `return` statements to handle different conditions. For example, here is a function that checks if a number is positive, negative, or zero and returns an appropriate message

```
def check_number(num):
  if num > 0:
    return "Positive"
  elif num < 0:
    return "Negative"
  else:
    return "Zero"
```

You can call this function with different numbers to see the results

```
print(check_number(10))   # Output: Positive
print(check_number(-5))   # Output: Negative
print(check_number(0))    # Output: Zero
```

```
Positive
Negative
Zero
```

Notice if we call as argument a variable that has been defined outside the function, it will automatically take that value (i.e., it's *global* to the function). If the variable is also defined inside, it will be *local* within the function.

```
global_var = 10

def edit_input(input_var):

    # Access the input variable
    print("Input you gave me", input_var)
```

```
    input_var = input_var + 5  # This modifies the local copy of
 ↪  input_var and not global_var
    print("Inside the function - modified input_var:", input_var)

    return input_var  # Return the modified value

def edit_global(input_var):

    global global_var # Make global_var accessible inside the
     ↪  function

    # Access the input variable
    print("Input you gave me", input_var)

    global_var = global_var + input_var  # This modifies the
 ↪  global variable
    print("Inside the function - modified global_var:",
     ↪  global_var)

    return None

# Call the function
edit_input(global_var)
print("Outside the function - global_var:", global_var)

# Call the function
edit_global(global_var)
print("Outside the function - global_var:", global_var)
```

```
Input you gave me 10
Inside the function - modified input_var: 15
Outside the function - global_var: 10
Input you gave me 10
Inside the function - modified global_var: 20
Outside the function - global_var: 20
```

Often times it is better to avoid global variables if possible, as they can lead to code that is harder to understand and maintain. Instead, prefer passing variables as arguments to functions and returning results. For example, if you would like to modify the value of `global_var`, you could simply assign the returned value of the function to it

```
global_var = edit_input(global_var)
print("Outside the function - global_var:", global_var)
```

```
Input you gave me 20
```

```
Inside the function - modified input_var: 25
Outside the function - global_var: 25
```

Functions can also have **default arguments**, which are used if no value is provided when the function is called. For example, here is a function that greets a user with a default name if none is provided

```python
def greet_with_default(name="Guest"):
  print(f"Hello, {name}!")

greet_with_default()
greet_with_default("Jesus")
```

```
Hello, Guest!
Hello, Jesus!
```

We used the same function, once without providing an argument (so it uses the default value "Guest") and once with a specific name ("Jesus").

We can also use **keyword arguments** to call functions. This allows us to specify the names of the parameters when calling the function, making it clear what each argument represents. For example

```python
def introduce(name, age):
  print(f"My name is {name} and I am {age} years old.")

introduce(name="Alba", age=30)
```

```
My name is Alba and I am 30 years old.
```

We can even change the order of the arguments when using keyword arguments, as shown above. You can also mix positional and keyword arguments, but positional arguments must come before keyword arguments.

```python
introduce("Alba", age=30) # This works
```

```
My name is Alba and I am 30 years old.
```

```python
#introduce(age=30, "Alba") # This will raise a SyntaxError
```

Positional arguments must be provided in the correct order, starting from the first parameter defined in the function. If you try to provide them in the wrong order, Python will raise a `TypeError`. For example, the following code will raise an error because the first argument is expected to be `name`, but we intended to provide an integer for `age`.

```python
#introduce(30, name="Alba") # This will raise a TypeError
```

Finally, note that the function needs to be defined before it is called in the code. If you try to call a function before its definition, Python will raise a `NameError`

indicating that the function is not defined.

```
#test_function()  # This will raise a NameError

def test_function():
  print("This is a test function.")
```

But the following will work correctly

```
def test_function():
  print("This is a test function.")

test_function()  # This will work correctly
```

```
This is a test function.
```

For this reason, function definitions are often placed at the beginning of a script or notebook cell, before any calls to those functions.

2.5.3 Conditional statements

Conditional statements allow you to control the flow of your program based on certain conditions. In Python, you can use `if`, `elif`, and `else` statements to execute different blocks of code depending on whether a condition is true or false. We have already seen an example of this in the `check_number` function above.

In the following example, the `do_something()` function will only be executed if `condition` evaluates to `True`, while `do_some_other_thing()` will always be executed.

```
if condition:
  do_something()

do_some_other_thing()
```

It is important to note that Python uses indentation to define the scope of code blocks. The code inside the `if` statement must be indented consistently to indicate that it belongs to that block.

```
a = 10

if a > 5:
  print("a is greater than 5")
  print("This line is also part of the if block")

print("This line is outside the if block")
```

```
a is greater than 5
This line is also part of the if block
```

```
This line is outside the if block
```

You can also nest `if` statements within each other to create more complex conditions. For example

```
a = 10

if a > 5:
  if a < 15:
    print("a is between 5 and 15")
  else:
    print("a is greater than or equal to 15")
else:
  print("a is less than or equal to 5")
```

```
a is between 5 and 15
```

Here, we first check if `a` is greater than `5`. If that condition is true, we then check if `a` is less than `15`. Depending on the outcome of these checks, different messages will be printed. Compared to the previous example, we also used an `else` statement to handle the case where `a` is not less than `15`.

We can also use `elif` (short for "else if") to check multiple conditions in a more concise way. For example

```
a = 10

if a < 5:
  print("a is less than 5")
elif a < 15:
  print("a is between 5 and 15")
else:
  print("a is greater than or equal to 15")
```

```
a is between 5 and 15
```

To reach the `elif` block, the first `if` condition must evaluate to `False`. If it evaluates to `True`, the code inside that block will be executed, and the rest of the conditions will be skipped. If none of the conditions are met, the code inside the `else` block will be executed.

Note that if statements can also be written in a single line using a ternary conditional operator. For example

```
a = 10
result = "a is greater than 5" if a > 5 else "a is less than or
 ↪  equal to 5"
print(result)
```

```
a is greater than 5
```

The above code assigns a different string to the variable `result` based on the condition `a > 5`. If the condition is true, it assigns "a is greater than 5"; otherwise, it assigns "a is less than or equal to 5".

2.5.4 Loops

Loops allow you to execute a block of code multiple times, which is useful for iterating over collections of data or performing repetitive tasks. In Python, there are two main types of loops: `for` loops and `while` loops.

`while` loops repeatedly execute a block of code as long as a specified condition is true. For example

```
count = 0
while count < 5:
  print("Count is", count)
  count += 1  # Increment count by 1

print("Final count is", count)
```

```
Count is 0
Count is 1
Count is 2
Count is 3
Count is 4
Final count is 5
```

In this example, the loop will continue to run as long as `count` is less than `5`. Inside the loop, we print the current value of `count` and then increment it by `1`. Once `count` reaches `5`, the condition becomes false, and the loop exits. Note that `count += 1` is a shorthand for `count = count + 1`.

`for` loops are used to iterate over a sequence (like a list, tuple, or string) or other iterable objects. We will see examples of such objects in the section on data structures. For the moment, let's look at a simple example of a `for` loop that iterates over a list of numbers

```
numbers = [1, 2, 3, 4, 5]
for num in numbers:
  print("Number is", num)
```

```
Number is 1
Number is 2
Number is 3
Number is 4
Number is 5
```

or alternatively, we can use the `range()` function to generate a sequence of numbers to iterate over

```python
for i in range(5):  # Generates numbers from 0 to 4
  print("i is:", i)
```

```
i is: 0
i is: 1
i is: 2
i is: 3
i is: 4
```

We can use the function `range` also to get a sequence of number to loop over. It follows the syntax `range(start, stop, step)`

- **Start**
  - Where the sequence starts: it includes the start value (first number will always be `start`)
  - Optional
  - Defaults to 0 (unless otherwise specified)
- **Stop**:
  - Where the sequence ends: it does not include the stop value (last number will always be `stop-step`)
  - Required field
- **Step**:
  - Step size of the sequence, i.e., how much we increase the value at each iteration: `start + step`, `start + 2*step`, `start + 3*step`, …
  - Optional
  - Defaults to 1 (unless otherwise specified)

```python
for i in range(2, 10, 2):  # Generates even numbers from 2 to 8
  print("i is", i)
```

```
i is 2
i is 4
i is 6
i is 8
```

But as mentioned before, `for` loops can iterate over any iterable object, not just sequences of numbers. For example, we can iterate over the characters in a string

```python
for letter in "Cemfi":
    print(letter)
```

```
C
e
m
f
i
```

or over a list of strings

```
months_of_year = ["January", "February", "March", "April", "May",
 ↪  "June", "July", "August", "September", "October", "November",
 ↪  "December"]

# Loop through the months and add some summer vibes
for month in months_of_year:
    if month == "June":
        print(f"Get ready to enjoy the summer break, it's
         ↪  {month}!")
    elif month == "July" or month =='August':
        print(f"{month} is perfect to find reasons to escape from
         ↪  Madrid")
    else:
        print(f"Winter is coming")
```

```
Winter is coming
Winter is coming
Winter is coming
Winter is coming
Winter is coming
Get ready to enjoy the summer break, it's June!
July is perfect to find reasons to escape from Madrid
August is perfect to find reasons to escape from Madrid
Winter is coming
Winter is coming
Winter is coming
Winter is coming
```

Where we combined loops with conditional statements to print different messages based on the current month.

Note that you can use the `break` statement to exit a loop prematurely when a certain condition is met, and the `continue` statement to skip the current iteration and move to the next one.

```
for i in range(10):
  if i == 5:
    break  # Exit the loop when i is 5
  print("i is", i)
```

```
i is 0
i is 1
i is 2
i is 3
i is 4
```

```
for i in range(10):
  if i % 2 == 0:
    continue  # Skip even numbers
  print("i is", i)
```

```
i is 1
i is 3
i is 5
i is 7
i is 9
```

You can also create nested loops, where one loop is placed inside another loop. This is useful for iterating over multi-dimensional data structures or performing more complex tasks.

```
for i in range(3):
  for j in range(2):
    print(f"i: {i}, j: {j}")
```

```
i: 0, j: 0
i: 0, j: 1
i: 1, j: 0
i: 1, j: 1
i: 2, j: 0
i: 2, j: 1
```

`enumerate()` is a built-in function that adds a counter to an iterable and returns it as an enumerate object. This is particularly useful when you need both the index and the value of items in a loop.

```
fruits = ["apple", "banana", "cherry"]
for index, fruit in enumerate(fruits):
    print(f"Index: {index}, Fruit: {fruit}")
```

```
Index: 0, Fruit: apple
Index: 1, Fruit: banana
Index: 2, Fruit: cherry
```

2.5.5 Exercises

Now that we have covered the basics of Python programming, it's time to practice what we've learned. Here are some exercises to help you reinforce your understanding of variables, data types, functions, conditionals, and loops.

1. Create two variables, `a` and `b`, and assign them the values `10` and `20`, respectively. Write a function that takes these two variables as input and returns their product and their difference.

2. Write a function called `is_even` that takes a number as input and returns `True` if the number is even and `False` otherwise. Try calling the function with different numbers to test it.
3. Write a loop that computes the result of the sum $\sum_{i=1}^{10} i^2$ and prints the result.
4. Write a loop to compute the product of all odd numbers between `1` and `20`. Print the final result. *Hint:* You could reuse the `is_even` function you defined earlier.
5. Compute the sum of all numbers between `1` and `1000` that are divisible by `3` or `5`. Print the final result.

2.6 Data Structures

The fundamental data types we have seen so far are useful for storing single values. However, in practice, we often need to work with collections of data. Python provides several built-in **collection types** to handle such cases. The most commonly used data structures in Python are

- **Lists**: Ordered, mutable collections of items
- **Tuples**: Ordered, immutable collections of items
- **Dictionaries**: Unordered, mutable collections of key-value pairs
- **Sets**: Unordered collections of unique items
- **Ranges**: Immutable sequences of numbers, often used for iteration

We will explore each of these types in more detail below.

2.6.1 Lists

We have already seen lists in some of the previous examples. A list is an ordered collection of items that can be of different types. Lists are mutable, meaning you can change their contents after creation. You can create a list by enclosing items in square brackets `[]`, separated by commas.

```
my_list = [1, 2.5, "Hello", True]
print(my_list)
```

```
[1, 2.5, 'Hello', True]
```

We can access individual elements in a list using their index, which starts at `0` for the first element. For example

```
first_element = my_list[0]
print("First element:", first_element)
```

```
First element: 1
```

You can also access elements from the end of the list using negative indices, where `-1` refers to the last element, `-2` to the second last, and so on.

```
last_element = my_list[-1]
print("Last element:", last_element)
```

```
Last element: True
```

Multiple elements can be accessed using slicing, which allows you to specify a range of indices. The syntax for slicing is `list[start:stop]`, where `start` is the index of the first element to include, and `stop` is the index of the first element to exclude.

```
sub_list = my_list[1:3]  # Elements at index 1 and 2
print("Sub-list:", sub_list)
```

```
Sub-list: [2.5, 'Hello']
```

Since lists are mutable, you can modify their contents. For example, you can change the value of an element at a specific index.

```
my_list[2] = "World"
print("After modification:", my_list)
```

```
After modification: [1, 2.5, 'World', True]
```

If we would like to add elements to a list, we can use the `append()` method to add an item to the end of the list or the `insert()` method to add an item at a specific index, or `extend()` to add multiple items at once.

```
my_list.append("New Item")
print("After appending:", my_list)
```

```
After appending: [1, 2.5, 'World', True, 'New Item']
```

```
my_list.insert(1, "Inserted Item")
print("After inserting:", my_list)
```

```
After inserting: [1, 'Inserted Item', 2.5, 'World', True, 'New Item']
```

```
my_list.extend([3, 4, 5])
print("After extending:", my_list)
```

```
After extending: [1, 'Inserted Item', 2.5, 'World', True, 'New Item', 3, 4, 5]
```

Note how these methods modify the original list in place without needing to reassign it, i.e., without needing to write `=`.

There are also options to remove items from a list. You can use the `remove()` method to remove the first occurrence of a specific value, the `pop()` method to remove an item at a specific index (or the last item if no index is provided), or the `clear()` method to remove all items from the list.

```
my_list.remove("World")
print("After removing 'World':", my_list)
```

```
After removing 'World': [1, 'Inserted Item', 2.5, True, 'New Item', 3, 4, 5]
```

```
popped_item = my_list.pop(2)  # Remove item at index 2
print("After popping index 2:", my_list)
print("Popped item:", popped_item)
```

```
After popping index 2: [1, 'Inserted Item', True, 'New Item', 3, 4, 5]
Popped item: 2.5
```

```
my_list.clear()
print("After clearing:", my_list)
```

```
After clearing: []
```

There is a convenient way to create lists using **list comprehensions**. List comprehensions provide a concise way to create lists based on existing iterables. The syntax is `[expression for item in iterable if condition]`, where `expression` is the value to be added to the list, `item` is the variable representing each element in the iterable, and `condition` is an optional filter. For example, here is how to create a list of squares of even numbers from `0` to `9`.

```
squares_of_even = [x**2 for x in range(10) if x % 2 == 0]
print("Squares of even numbers:", squares_of_even)
```

```
Squares of even numbers: [0, 4, 16, 36, 64]
```

Let's break down the list comprehension above:

- `x**2`: This is the expression that defines what each element in the new list will be. In this case, it's the square of `x`.
- `for x in range(10)`: This part iterates over the numbers from `0` to `9`.
- `if x % 2 == 0`: This is a condition that filters the numbers, including only even numbers in the new list. It uses the modulus operator `%` to check if `x` is divisible by `2`. If a number is divisible by `2`, the remainder is `0`, indicating that it is even.

2.6.2 Tuples

Tuples are similar to lists in that they are ordered collections of items. However, tuples are immutable, meaning that once they are created, their contents cannot be changed. You can create a tuple by enclosing items in parentheses `()`, separated by commas.

```
my_tuple = (1, 2.5, "Hello", True)
print(my_tuple)
```

```
(1, 2.5, 'Hello', True)
```

You can access elements in a tuple using indexing and slicing, just like with lists.

```
first_element = my_tuple[0]
print("First element:", first_element)
second_element = my_tuple[1]
print("Second element:", second_element)
last_element = my_tuple[-1]
print("Last element:", last_element)
sub_tuple = my_tuple[1:3]  # Elements at index 1 and 2
print("Sub-tuple:", sub_tuple)
```

```
First element: 1
Second element: 2.5
Last element: True
Sub-tuple: (2.5, 'Hello')
```

Note that we have seen tuples before when we defined functions that return multiple values. In such cases, Python automatically packs the returned values into a tuple, which can then be unpacked into separate variables.

```
def get_coordinates():
  x = 10
  y = 20
  return x, y  # Returns a tuple (10, 20)

x_coord, y_coord = get_coordinates()  # Unpacks the tuple into
 ↪  separate variables
print("X coordinate:", x_coord)
print("Y coordinate:", y_coord)
```

```
X coordinate: 10
Y coordinate: 20
```

Note that tuples are faster than lists for certain operations due to their immutability, making them a good choice for storing data that should not change. If you need to be able to modify the contents, use a list instead. For example, the following code will raise an error because we are trying to change an element of a tuple

```
#my_tuple[1] = 3.0  # This will raise a TypeError
```

While tuples are immutable, you can concatenate two tuples to create a new tuple

```
tuple1 = (1, 2, 3)
tuple2 = (4, 5, 6)
combined_tuple = tuple1 + tuple2
```

```
print("Combined tuple:", combined_tuple)
```

```
Combined tuple: (1, 2, 3, 4, 5, 6)
```

or you can repeat a tuple multiple times

```
repeated_tuple = tuple1 * 3
print("Repeated tuple:", repeated_tuple)
```

```
Repeated tuple: (1, 2, 3, 1, 2, 3, 1, 2, 3)
```

Unpacking can also be used with tuples. For example, you can unpack the elements of a tuple into separate variables

```
my_tuple = (10, 20, 30)
a, b, c = my_tuple
print("a:", a)
print("b:", b)
print("c:", c)
```

```
a: 10
b: 20
c: 30
```

If you don't want to unpack all elements, you can use the asterisk (`*`) operator to capture the remaining elements in a list

```
my_tuple = (10, 20, 30, 40, 50)
a, b, *rest = my_tuple
print("a:", a)
print("b:", b)
print("rest:", rest)
```

```
a: 10
b: 20
rest: [30, 40, 50]
```

It is also common to use `_` (underscore) as a variable name for values that you want to ignore during unpacking

```
my_tuple = (10, 20, 30)
a, _, c = my_tuple  # Ignore the second element
print("a:", a)
print("c:", c)
```

```
a: 10
c: 30
```

2.6.3 Dictionaries

Dictionaries are unordered collections of key-value pairs. Each key is unique and is used to access its corresponding value. Dictionaries are mutable, meaning you can change their contents after creation. The keys in a dictionary must be unique and immutable (e.g., strings, numbers, or tuples), while the values can be of any data type and can be duplicated. You can create a dictionary by enclosing key-value pairs in curly braces `{}`, with each key and value separated by a colon `:` and pairs separated by commas.

```
my_dict = {"name": "Alba", "age": 30, "is_student": False}
print(my_dict)
```

```
{'name': 'Alba', 'age': 30, 'is_student': False}
```

You can access values in a dictionary using their keys. For example

```
name = my_dict["name"]
print("Name:", name)
```

```
Name: Alba
```

You can also add new key-value pairs or update existing ones

```
my_dict["city"] = "Madrid"  # Add a new key-value pair
print("After adding city:", my_dict)
```

```
After adding city: {'name': 'Alba', 'age': 30, 'is_student': False, 'city': 'Madrid'}
```

Alternatively, you can use the `update()` method to add or update multiple key-value pairs at once

```
my_dict.update({"age": 31, "country": "Spain"})
print("After updating age and adding country:", my_dict)
```

```
After updating age and adding country: {'name': 'Alba', 'age': 31, 'is_student': False, 'city': '
```

Note that if you use a key that already exists in the dictionary, the corresponding value will be updated. This applies whether you use the assignment syntax or the `update()` method.

The keys and values can be accessed using the `keys()` and `values()` methods, respectively. You can also use the `items()` method to get a list of key-value pairs as tuples.

```
keys = my_dict.keys()
print("Keys:", keys)
```

```
Keys: dict_keys(['name', 'age', 'is_student', 'city', 'country'])
```

```
values = my_dict.values()
print("Values:", values)
```

```
Values: dict_values(['Alba', 31, False, 'Madrid', 'Spain'])
```

```
items = my_dict.items()
print("Items:", items)
```

```
Items: dict_items([('name', 'Alba'), ('age', 31), ('is_student', False), ('city', 'Madr
```

The latter is particularly useful for iterating over both keys and values in a loop.

We can remove key-value pairs from a dictionary using the `del` statement or the `pop()` method.

```
del my_dict["is_student"]
print("After deleting is_student:", my_dict)
```

```
After deleting is_student: {'name': 'Alba', 'age': 31, 'city': 'Madrid', 'country': 'Sp
```

```
age = my_dict.pop("age")
print("After popping age:", my_dict)
print("Popped age:", age)
```

```
After popping age: {'name': 'Alba', 'city': 'Madrid', 'country': 'Spain'}
Popped age: 31
```

2.6.4 Sets

Sets are unordered collections of unique items. They are mutable, meaning you can change their contents after creation. Sets are useful for storing items when the order does not matter and duplicates are not allowed. You can create a set by enclosing items in curly braces `{}`, separated by commas.

```
my_set = {1, 2, 3, 4, 5}
print("Set:", my_set)
```

```
Set: {1, 2, 3, 4, 5}
```

You can also create a set from an iterable, such as a list, using the `set()` constructor.

```
my_list = [1, 2, 2, 3, 4, 4, 5]
my_set_from_list = set(my_list)
print("Set from list:", my_set_from_list)
```

```
Set from list: {1, 2, 3, 4, 5}
```

You can add items to a set using the `add()` method and remove items using the `remove()` or `discard()` methods.

```
my_set.add(6)
print("After adding 6:", my_set)
```

```
After adding 6: {1, 2, 3, 4, 5, 6}
```

```
my_set.remove(3)
print("After removing 3:", my_set)
```

```
After removing 3: {1, 2, 4, 5, 6}
```

```
my_set.discard(10)  # Does not raise an error if 10 is not in the
 ↪  set
print("After discarding 10:", my_set)
```

```
After discarding 10: {1, 2, 4, 5, 6}
```

There is also a `frozenset` type, which is an immutable version of a set. Once created, the contents of a frozenset cannot be changed. You can create a frozenset using the `frozenset()` constructor.

```
my_frozenset = frozenset([1, 2, 3, 4, 5])
print("Frozenset:", my_frozenset)
```

```
Frozenset: frozenset({1, 2, 3, 4, 5})
```

Sets are particularly useful for performing mathematical set operations such as union, intersection, difference, and symmetric difference. For example

```
set_a = {1, 2, 3, 4}
set_b = {3, 4, 5, 6}
union_set = set_a.union(set_b)
print("Union:", union_set)
```

```
Union: {1, 2, 3, 4, 5, 6}
```

```
intersection_set = set_a.intersection(set_b)
print("Intersection:", intersection_set)
```

```
Intersection: {3, 4}
```

```
difference_set = set_a.difference(set_b)
print("Difference (A - B):", difference_set)
```

```
Difference (A - B): {1, 2}
```

```
symmetric_difference_set = set_a.symmetric_difference(set_b)
print("Symmetric Difference:", symmetric_difference_set)
```

```
Symmetric Difference: {1, 2, 5, 6}
```

2.6.5 Ranges

Ranges are immutable sequences of numbers, commonly used for iteration in loops. You can create a range using the `range()` function, which generates a sequence of numbers based on the specified start, stop, and step values. The syntax is `range(start, stop, step)`, where `start` is the first number in the sequence (inclusive), `stop` is the end of the sequence (exclusive), and `step` is the increment between each number.

```
my_range = range(0, 10, 2)  # Generates numbers from 0 to 8 with
 ↪  a step of 2
print("Range:", list(my_range)) # Convert to list for display
```

```
Range: [0, 2, 4, 6, 8]
```

You can also create a range with just the `stop` value, in which case the sequence starts from `0` and increments by `1` by default.

```
my_range_default = range(5)  # Generates numbers from 0 to 4
print("Range with default start and step:",
 ↪  list(my_range_default))
```

```
Range with default start and step: [0, 1, 2, 3, 4]
```

You have seen earlier how to use ranges in `for` loops to iterate over a sequence of numbers. Ranges are memory efficient because they generate numbers on-the-fly and do not store the entire sequence in memory, making them suitable for large sequences.

2.6.6 Mutable vs. Immutable Objects

In the examples up until now you have already seen that data types can be classified as either **mutable** or **immutable** based on whether their values can be changed after they are created.

- **Mutable objects**: These objects can be modified after they are created. Examples of mutable data types in Python include lists, dictionaries, and sets. When you modify a mutable object, you are changing the object itself, and any other references to that object will reflect the changes.
- **Immutable objects**: These objects cannot be modified after they are created. Examples of immutable data types in Python include integers, floats, strings, and tuples. When you attempt to modify an immutable object, you are actually creating a new object with the modified value, leaving the original object unchanged.

An important implication of mutability is what happens when you assign one variable to another. For mutable objects, both variables will reference the same object in memory, so changes made through one variable will affect the other. For immutable objects, each variable will reference its own separate object.

```
# Mutable example with lists
list1 = [1, 2, 3]
list2 = list1  # Both variables reference the same list
list2.append(4)  # Modify list2
print("list1:", list1)  # list1 is also affected
print("list2:", list2)
```

```
list1: [1, 2, 3, 4]
list2: [1, 2, 3, 4]
```

```
# Immutable example with strings
str1 = "Hello"
str2 = str1  # Both variables reference the same string
str2 += ", World!"  # Modify str2 (creates a new string)
print("str1:", str1)  # str1 remains unchanged
print("str2:", str2)
```

```
str1: Hello
str2: Hello, World!
```

The concept of mutability is important to understand when working with data structures and functions in Python, as it can affect how data is passed and modified within your code. When passing mutable objects to functions, changes made to the object within the function will affect the original object outside the function.

```
def modify_list(input_list):
    input_list.append(100)  # Modifies the original list

my_list = [1, 2, 3]
modify_list(my_list)
print(my_list)  # my_list is changed
```

```
[1, 2, 3, 100]
```

In contrast, passing immutable objects to functions will not affect the original object.

```
def modify_int(input_int):
    input_int += 10  # Creates a new integer

my_int = 5
modify_int(my_int)
print(my_int)  # my_int remains unchanged
```

```
5
```

Therefore, it is crucial to be aware of the mutability of the data types you are working with to avoid unintended side effects in your code.

2.6.7 Exercises

Now that we have covered the basics of data structures in Python, it's time to practice what we've learned. Here are some exercises to help you reinforce your understanding of lists, tuples, dictionaries, sets, and ranges.

1. Create a list of the first 10 square numbers (i.e., 1, 4, 9, ..., 100) using a list comprehension. Print the resulting list.
2. Create a tuple containing the names of the days of the week. Access and print the name of the third day.
3. Create a dictionary that maps the names of three countries to their respective capitals. Access and print the capital of one of the countries.
4. Create a set containing the unique vowels in the word "programming". Print the resulting set.
5. Create a range of numbers from 1 to 20 with a step of 3. Use a for loop to iterate over the range and print each number.

2.7 Object-Oriented Programming (OOP) Basics

Object-Oriented Programming (OOP) is a programming paradigm that organizes code around "objects" - which combine data (attributes) and functions (methods) that operate on that data. Think of objects as self-contained units that represent real-world entities or concepts. In Python, everything is an object, including basic data types like integers and strings. Therefore, we have been using OOP concepts all along without being explicit about it.

2.7.1 Classes and Objects

A **class** is like a blueprint or template for creating objects. An **object** is a specific instance created from that class. For example, if "Car" is a class, then "my_toyota" and "your_honda" would be objects (instances) of that class.

Here's a simple example of defining a class and creating objects from it

```
# Define a class
class BankAccount:
    def __init__(self, owner, balance=0):
        self.owner = owner
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print(f"Deposited ${amount}. New balance:
         ↪  ${self.balance}")

    def withdraw(self, amount):
        if amount > self.balance:
```

```
            print("Insufficient funds!")
        else:
            self.balance -= amount
            print(f"Withdrew ${amount}. New balance:
             ↪  ${self.balance}")

# Create objects (instances)
account1 = BankAccount("Alba", 1000)
account2 = BankAccount("Jesus", 500)

# Use methods
account1.deposit(200)
account1.withdraw(300)

# Check balances (accessing attributes)
print(f"{account1.owner}'s balance: ${account1.balance}")
print(f"{account2.owner}'s balance: ${account2.balance}")
```

```
Deposited $200. New balance: $1200
Withdrew $300. New balance: $900
Alba's balance: $900
Jesus's balance: $500
```

The `__init__` method is a special method called a **constructor** that runs automatically when you create a new object. The `self` parameter refers to the instance itself and is used to access its attributes and methods.

2.7.2 Attributes and Methods

Attributes are variables that belong to an object and store its data. **Methods** are functions that belong to an object and define its behavior.

```
class Student:
    def __init__(self, name, student_id):
        self.name = name              # attribute
        self.student_id = student_id  # attribute
        self.courses = []             # attribute

    def enroll(self, course):         # method
        self.courses.append(course)
        print(f"{self.name} enrolled in {course}")

    def get_courses(self):            # method
        return self.courses

# Create and use a student object
student = Student("Alba", "S12345")
```

```
student.enroll("Artificial Intelligence and Big Data")
student.enroll("Python Programming")
print(f"{student.name}'s courses: {student.get_courses()}")
```

```
Alba enrolled in Artificial Intelligence and Big Data
Alba enrolled in Python Programming
Alba's courses: ['Artificial Intelligence and Big Data', 'Python Programming']
```

2.7.3 Why Use OOP?

OOP helps organize complex programs by grouping related data and functionality together. This makes code:

- **More intuitive**: Objects model real-world entities
- **Easier to maintain**: Changes to one class don't affect unrelated code
- **Reusable**: Classes can be used in multiple parts of your program

In data science, you'll often work with objects like DataFrames (from pandas), models (from scikit-learn), or plots (from matplotlib), even if you don't create your own classes frequently.

```
# Example: You're already using OOP when working with lists!
my_list = [1, 2, 3]      # my_list is an object of class 'list'
my_list.append(4)        # append is a method
my_list.sort()           # sort is a method
print(len(my_list))      # len works with the object's internal
 ↪  data
```

```
4
```

For **this course**, understanding how to use objects and their methods is more important than creating complex class hierarchies. Most of the time, you'll be **using classes created by others** (like pandas DataFrames or scikit-learn models) **rather than writing your own**.

2.8 Essential Packages

In this section, we will introduce some of the most essential packages in Python for data science and scientific computing. These packages provide powerful tools and functionalities that make it easier to work with data, perform numerical computations, and create visualizations.

> **ℹ Modules vs. Packages**
>
> A module, in Python, is a program that can be imported into interactive mode or other programs for use. A Python package typically comprises

multiple modules. Physically, a package is a directory containing modules and possibly subdirectories, each potentially containing further modules. Conceptually, a package links all modules together using the package name for reference.

2.8.1 Scientific Computing: NumPy

NumPy (**Num**erical **Py**thon) is one of the most common packages used in Python. In fact, numerous computational packages that offer scientific capabilities utilize NumPy's array objects as a standard interface for data exchange. That's why understanding NumPy arrays and array-based computing principles can save your time in the future.

NumPy offers a vast array of efficient methods for creating and manipulating numerical data arrays. Unlike Python lists, which can accommodate various data types within a single list, NumPy arrays require homogeneity among their elements for efficient mathematical operations. Utilizing NumPy arrays provides advantages such as faster execution and reduced memory consumption compared to Python lists. With NumPy, data storage is optimized through the specification of data types, enhancing code optimization.

> **Note**
>
> Documentation for this package is available at https://numpy.org/doc/stable/.

To use NumPy, in your code, you typically import it with the alias `np`

```
import numpy as np
```

2.8.1.1 Creating NumPy Arrays

Arrays serve as a fundamental data structure within the NumPy. They represen a grid of values containing information on raw data, element location, and interpretation. Elements share a common data type, known as the array dtype.

One method of initializing NumPy arrays involves using Python lists, with nested lists employed for two- or higher-dimensional data structures.

```
a = np.array([1, 2, 3, 4, 5, 6])
print("1D array:", a)
```

```
1D array: [1 2 3 4 5 6]
```

We can access the elements through indexing.

```
a[0]
```

```
np.int64(1)
```

Arrays are N-Dimensional (that's why sometimes we refer to them as NDarray). That means that NumPy arrays will encompass vector (1-Dimensions), Matrices (2D) or tensors (3D and higher). We can get all the information of the array by checking its attributes. To create a 2D array, we can use nested lists:

```
a = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])
```

Mathetematically, we can think of this as a matrix with 2 rows and 4 columns, i.e.,

$$a = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

We can check its attributes to get more information about the array:

```
print('Dimensions/axes:', a.ndim)
print('Shape (size of array in each dimension):', a.shape)
print('Size (total number of elements):', a.size)
print('Number of bytes:', a.nbytes)
print('Data type:', a.dtype)
print('Item size (in bytes):', a.itemsize)
```

```
Dimensions/axes: 2
Shape (size of array in each dimension): (2, 4)
Size (total number of elements): 8
Number of bytes: 64
Data type: int64
Item size (in bytes): 8
```

We have already seen how to access elements in a 1D array. For 2D arrays, we can use two indices: the first for the row and the second for the column.

```
element = a[0, 2]  # Access the element in the first row and
 ↪  third column
print("Element at (0, 2):", element)
```

```
Element at (0, 2): 3
```

We can also use slicing to access subarrays. For example, to get the first two rows and the first three columns:

```
subarray = a[0:2, 0:3]
print("Subarray:\n", subarray)
```

```
Subarray:
 [[1 2 3]
 [5 6 7]]
```

We don't need to specify both indices all the time. For example, to get the first row, we can do

```
first_row = a[0, :]
print("First row:", first_row)
```

```
First row: [1 2 3 4]
```

or to get the second column

```
second_column = a[:, 1]
print("Second column:", second_column)
```

```
Second column: [2 6]
```

We can initialize arrays using different commands depending on what we aim.

For instance, the most straightforward case would be to pass a list to `np.array()` to create one:

```
arr1 = np.array([5,6,7])
arr1
```

```
array([5, 6, 7])
```

However, sometimes we are more ambiguous and have no information on what our array contains. We just need to be able to initialize an array so that later on, our code, can update it. For this, we typically create arrays of the desired dimensions and fill them with zeros (`np.zeros()`), ones (`np.ones()`), with a given value (`np.full()`) or with without initializing (`np.empty()`).

> **Tip**
>
> When working with large data, `np.empty()` can be faster and more efficient. Also, large arrays can take up most of your memory and, in those cases, carefully establishing the `dtype()` can help to manage memory more efficiently (i.e., chose 8 bits over 64 bits.)

```
np.zeros(4)
```

```
array([0., 0., 0., 0.])
```

```
np.ones((2,3))
```

```
array([[1., 1., 1.],
       [1., 1., 1.]])
```

To create higher-dimensional arrays, we can pass a tuple representing the shape of the array:

```
np.ones((3,2,1))
```

```
array([[[1.],
        [1.]],

       [[1.],
        [1.]],

       [[1.],
        [1.]]])
```

This created a 3D array with 3 rows, 2 columns and a depth of 1.

We can use `np.full()` to create an array of constant vales that we specify in the `fill_value` option.

```
np.full((2,2,) , fill_value= 4)
```

```
array([[4, 4],
       [4, 4]])
```

`np.empty()` creates an array without initializing its values. The values in the array will be whatever is already present in the allocated memory, which can be random and unpredictable.

```
np.empty(2)
```

```
array([4.94737384e+086, 8.17811799e+191])
```

With `np.linspace()`, we can create arrays with evenly spaced values over a specified range. The syntax is `np.linspace(start, stop, num)`, where `start` is the starting value, `stop` is the ending value, and `num` is the number of evenly spaced values to generate.

```
np.linspace(0, 1, 5)  # Generates 5 evenly spaced values between
 ↪  0 and 1
```

```
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
```

`np.arange()` is another useful function to create arrays with evenly spaced values, similar to the built-in `range()` function but returning a NumPy array. The syntax is `np.arange(start, stop, step)`, where `start` is the starting value, `stop` is the ending value (exclusive), and `step` is the increment between each value.

```
np.arange(0, 10, 2)  # Generates values from 0 to 8 with a step
 ↪  of 2
```

```
array([0, 2, 4, 6, 8])
```

Note that both `np.linspace()` and `np.arange()` can be used to create sequences of numbers, but they differ in how you specify the spacing and the number of elements. In general, use `np.linspace()` when you want a specific number of evenly spaced values over a range, and use `np.arange()` when you want to specify the step size between values.

Sometimes, you might also need to create identity matrices, which are square matrices with ones on the diagonal and zeros elsewhere. You can use `np.eye()` to create an identity matrix of a specified size.

```
np.eye(3)  # Creates a 3x3 identity matrix
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

Or you might want to create diagonal matrices with specific values on the diagonal. You can use `np.diag()` for this purpose.

```
np.diag([1, 2, 3])  # Creates a diagonal matrix with 1, 2, 3 on
 ↪  the diagonal
```

```
array([[1, 0, 0],
       [0, 2, 0],
       [0, 0, 3]])
```

Finally, to create random arrays, NumPy provides several functions in the `np.random` module. For example, you can create an array of random floats between 0 and 1 using `np.random.rand()`, or an array of random integers within a specified range using `np.random.randint()`, or a normal distribution using `np.random.randn()`.

```
np.random.rand(2, 3)  # Creates a 2x3 array of random floats
 ↪  between 0 and 1
```

```
array([[0.68505918, 0.58305047, 0.22019802],
       [0.77208186, 0.50208612, 0.08031959]])
```

```
np.random.randint(0, 10, size=(2, 3))  # Creates a 2x3 array of
 ↪  random integers between 0 and 9
```

```
array([[3, 1, 7],
       [1, 6, 6]])
```

```
np.random.randn(2, 3)  # Creates a 2x3 array of random floats
 ↪  from a standard normal distribution
```

```
array([[-0.59439944,  0.55143246,  0.32346888],
```

```
       [-0.65271616, -0.43510666, -0.88092486]])
```

> **Random Seed**
>
> When generating random numbers, it's often useful to set a random seed using `np.random.seed()`. This ensures that the sequence of random numbers generated is reproducible, meaning that you will get the same random numbers each time you run your code with the same seed. This is particularly important for debugging and sharing results.

2.8.1.2 Managing Array Data

Arrays accept common operations like sorting, concatenating and finding unique elements.

For instance, using the `sort()` method we can sort elements within an array.

```
arr1 = np.array((10,2,5,3,50,0))
np.sort(arr1)
```

```
array([ 0,  2,  3,  5, 10, 50])
```

In multidimensional arrays, we can sort the elements of a given dimension by specifying the axis (0 across rows, 1 across columns)

```
mat1 = np.array([[1,2,3],[8,1,5]])
mat1
```

```
array([[1, 2, 3],
       [8, 1, 5]])
```

```
mat1.sort(axis=1)  # Sort across columns
mat1
```

```
array([[1, 2, 3],
       [1, 5, 8]])
```

Using `concatenate` we can join the elements of two arrays along an existing axis.

```
arr1 = np.array((1,2,3))
arr2 = np.array((6,7,8))
np.concatenate((arr1,arr2))
```

```
array([1, 2, 3, 6, 7, 8])
```

Instead, if we want to concatenate along a new axis, we sue `vstack()` and `hstack()`

```
np.vstack((arr1,arr2))  # Vertical stack
```

```
array([[1, 2, 3],
       [6, 7, 8]])
```

```
np.hstack((arr1,arr2))  # Horizontal stack
```

```
array([1, 2, 3, 6, 7, 8])
```

It is also possible to reshape arrays. For instance, let's reshape the concatenation of `arr1` and `arr2` to 3 rows and 2 columns

```
arr_c = np.concatenate((arr1,arr2))
arr_c.reshape((3,2))
```

```
array([[1, 2],
       [3, 6],
       [7, 8]])
```

We can also perform aggregation functions over all elements, like finding the minimum, maximum, means, sum of elements and much more.

```
print(arr1.min())
print(arr1.sum())
print(arr1.max())
print(arr1.mean())
```

```
1
6
3
2.0
```

This can also be done over a specific axis in multidimensional arrays. For example, let's create a 2D array and find the sum across rows and columns

```
mat2 = np.array([[1,2,3],[4,5,6]])
print(mat2.sum(axis=0))  # Sum across rows
print(mat2.sum(axis=1))  # Sum across columns
```

```
[5 7 9]
[ 6 15]
```

It is also possible to get only the unique elements of an array or to count how many elements are repeated.

```
arr1 =np.array((1,2,3,3,1,1,5,6,7,8,11,11))
print(np.unique(arr1))
unq, count = np.unique(arr1, return_counts=True)
print("Unique elements:", unq)
print("Counts:", count)
```

```
[ 1  2  3  5  6  7  8 11]
Unique elements: [ 1  2  3  5  6  7  8 11]
Counts: [3 1 2 1 1 1 1 2]
```

Using `where()`, we can find the indices of elements that satisfy a given condition.

```
arr1 = np.array((10,15,20,25,30,35,40))
indices = np.where(arr1 > 25)
print("Indices of elements greater than 25:", indices)
```

```
Indices of elements greater than 25: (array([4, 5, 6]),)
```

We can also use boolean indexing to filter elements based on a condition.

```
filtered_elements = arr1[arr1 > 25]
print("Elements greater than 25:", filtered_elements)
```

```
Elements greater than 25: [30 35 40]
```

And we can replace elements that meet a condition using `np.where()`

```
new_arr = np.where(arr1 > 25, -1, arr1)  # Replace elements
 ↪  greater than 25 with -1
print("Array after replacement:", new_arr)
```

```
Array after replacement: [10 15 20 25 -1 -1 -1]
```

2.8.1.3 Array Operations

NumPy arrays support common operations as addition, substraction and multiplication. These operations are performed element-wise, meaning that they are applied to each corresponding element in the arrays.

```
A = np.array(((1,2,3),
              (4,5,6)))
B = np.array(((10,20,30),
              (40,50,60)))
```

Element-wise addition, substraction and multiplication can be performed with `+`, `-` and `*`.

```
A + B
```

```
array([[11, 22, 33],
       [44, 55, 66]])
```

```
B - A
```

```
array([[ 9, 18, 27],
```

```
       [36, 45, 54]])
```

```
A * B
```

```
array([[ 10,  40,  90],
       [160, 250, 360]])
```

To multiply (`*`) or divide (`/`) all elements by an scalar, we just specify the scalar.

```
A * 10
```

```
array([[10, 20, 30],
       [40, 50, 60]])
```

```
B / 10
```

```
array([[1., 2., 3.],
       [4., 5., 6.]])
```

Note that NumPy automatically **broadcasts** the scalar to all elements of the array.

Broadcasting

Broadcasting is a powerful mechanism in NumPy that allows operations to be performed on arrays of different shapes. When performing operations between arrays of different shapes, NumPy automatically expands the smaller array along the dimensions of the larger array so that they have compatible shapes. This process is called broadcasting.
For example, consider adding a 1D array to a 2D array. NumPy will "broadcast" the 1D array across the rows of the 2D array to perform the addition.

```
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = np.array([10, 20, 30])  # 1D array
C = A + B  # B is broadcasted across the rows of A
print(C)
```

```
[[11 22 33]
 [14 25 36]]
```

Comparing NumPy arrays is also possible using operators as `==`, `!=`, and the like. Comparisons will result in an array of booleans indicating if the condition is met for a given element.

```
arr1 = np.array(((1,2,3),(4,5,6)))
arr2 = np.array(((1,5,3),(7,2,6)))
arr1==arr2
```

```
array([[ True, False,  True],
       [False, False,  True]])
```

Recall that we use double equals `==` for comparison, while a single equals `=` is used for assignment.

Note that elemen-wise multiplication is different from matrix multiplication. Matrix multiplication is achieved with either `@` or `matmul()`.

```
np.matmul(arr1,arr2.T) # Note the transpose of arr2 to match
 ↪  dimensions
```

```
array([[20, 29],
       [47, 74]])
```

2.8.1.4 Exercises

1. Create a 1D array with all integer elements from 1 to 10 (both included). No hard-coding allowed!
2. Form array you created in 1, create one that contains all odd elements and one with all even elements.
3. Create a new array that replaces all elements in 1 that are odd by -1.
4. Create a 3-by-3 matrix filled with 'True' values (i.e., booleans).
5. Suppose you have array `a=np.array(['a','b','c','d','e','f','g'])` and `b = np.array(['g','h','c','a','e','w','g'])`. Find all elements that are equal. Can you get the position where the elements of both arrays match?
6. Write a function that takes a element an array and prints elements that are divisible by a given number. Try it creating an array from 1 to 20 and printiing divisibles by 3.

7. Consider two matrices, A and B, both of size 100x100, filled with random integer values between 1 and 10. Implement a function to perform element-wise multiplication of these matrices using nested loops. Implement the same operation using Numpy's vectorized multiplication. Repeart againnwith matrices of size 1000x1000, 10000x10000, 1000000x1000000 and compare the execution time. Which one is faster?

2.8.2 Data Management: Pandas

Pandas is a an open source library providing high-performance, easy-to-use data structures and data analysis tools. Pandas is particularly suited to the analysis of *tabular* data, i.e. data that can can go into a table. In other words, if you can imagine the data in an Excel spreadsheet, then Pandas is the tool for the job.

- A fast and efficient DataFrame object for data manipulation with indexing
- Tools for reading and writing data: CSV and text files, Excel, SQL
- Intelligent data alignment and integrated handling of missing data

- Flexible reshaping and pivoting of data sets
- Intelligent label-based slicing, indexing, and subsetting of large data sets
- High performance aggregating, merging, joining or transforming data
- Hierarchical indexing provides an intuitive way of working with high-dimensional data
- Time series-functionality: date-based indexing, frequency conversion, moving windows, date shifting and lagging

> **Note**
>
> Documentation for this package is available at https://pandas.pydata.org/docs/.

To use Pandas, you typically import it with the alias `pd`

```
import pandas as pd
```

Pandas builds on two main data structures: `Series` and `DataFrames`. `Series` represent 1D arrays while `DataFrames` are 2D labeled arrays. The easiest way to think about both structures is to conceptualize `DataFrames` as containers of lower dimension data. That is, `DataFrames` columns are composed of `Series`, and each of the elements of a `Series` (i.e., the rows of the `DataFrame`) are individual scalar (numbers or strings) values. In plain words, `Series` are columns made of scalar elements and `DataFrames` are collections of `Series` that get an assigned label. All pandas data structures are value-mutable (i.e., we can change the values of elements and replace `DataFrames`) but some are not always size-mutable. The length of a Series cannot be changed, but, for example, columns can be inserted into a DataFrame.

2.8.2.1 Pandas Series

A `Series` is a one-dimensional labeled array capable of holding any data type (integers, strings, floating point numbers, Python objects, etc.). The axis labels are collectively referred to as the index. A `Series` can be created from a list, dictionary, or scalar value using the `pd.Series()` constructor. To create a `Series` from a list, you can do the following:

```
data = [10, 20, 30, 40, 50]
series = pd.Series(data)
```

If you want to specify custom index labels, you can pass a list of labels to the `index` parameter:

```
data = [10, 20, 30, 40, 50]
labels = ['a', 'b', 'c', 'd', 'e']
series = pd.Series(data, index=labels)
```

You can additionally assigne a name to the `Series` using the `name` parameter:

```
data = [10, 20, 30, 40, 50]
labels = ['a', 'b', 'c', 'd', 'e']
series = pd.Series(data, index=labels, name='My Series')
```

These functions work the same way when creating a `Series` from a NumPy array. When creating a `Series` from a dictionary, the keys of the dictionary become the index labels, and the values become the data:

```
data = {'a': 10, 'b': 20, 'c': 30}
series = pd.Series(data)
```

You can access elements in a `Series` using their index labels or integer positions. For example, to access the element with label 'b':

```
value = series['b']
print("Value at index 'b':", value)
```

```
Value at index 'b': 20
```

If you want to access elements by their integer position, you can use the `iloc` attribute:

```
value = series.iloc[1]  # Access the second element (index 1)
print("Value at position 1:", value)
```

```
Value at position 1: 20
```

Note that both label-based and positional indexing can be used interchangeably in many cases.

`.loc` is used for label-based indexing, which means you access elements by their index labels:

Syntax	Description	Example	Result
`series.loc[label]`	Single label access	`s.loc['b']`	Value at index 'b'
`series.loc[label_list]`	Multiple labels	`s.loc[['a', 'c']]`	Series with values at 'a' and 'c'
`series.loc[start:end]`	Slicing by labels (inclusive)	`s.loc['a':'c']`	Series from 'a' to 'c' (inclusive)
`series.loc[condition]`	Boolean indexing	`s.loc[s > 5]`	Values where condition is True

`.iloc` is used for positional indexing, which means you access elements by their integer position in the Series:

Syntax	Description	Example	Result
`series.iloc[position]`	Single position access	`s.iloc[1]`	Value at position 1
`series.iloc[positions_list]`	Multiple positions	`s.iloc[[0, 2]]`	Series with values at positions 0 and 2
`series.iloc[start:end]`	Slice by positions (exclusive end)	`s.iloc[1:3]`	Series from position 1 to 2
`series.iloc[negative_position]`	Negative indexing	`s.iloc[-1]`	Value at last position

Key Differences:

1. Indexing method:
   - `.loc` uses the actual index labels (strings, dates, etc.)
   - `.iloc` uses integer positions (0, 1, 2, …)
2. Slicing behavior:
   - `.loc` slicing is inclusive of both endpoints
   - `.iloc` slicing is exclusive of the end position

You can retrieve all index labels and values of a `Series` using the `index` and `values` attributes, respectively:

```
index_labels = series.index
print("Index labels:", index_labels)
```

```
Index labels: Index(['a', 'b', 'c'], dtype='object')
```

```
values = series.values
print("Values:", values)
```

```
Values: [10 20 30]
```

You can perform various operations on `Series`, such as arithmetic operations, aggregation functions, and filtering. For example, to add a scalar value to all elements in the `Series`:

```
new_series = series + 5
print("Series after adding 5:\n", new_series)
```

```
Series after adding 5:
 a    15
b    25
```

```
c    35
dtype: int64
```

You can also filter the `Series` based on a condition:

```
filtered_series = series[series > 20]
print("Filtered Series (values > 20):\n", filtered_series)
```

```
Filtered Series (values > 20):
 c    30
dtype: int64
```

They work and behave similarly to NumPy arrays in many ways but with additional functionality for handling missing data and labeled data.

2.8.2.2 Pandas DataFrames

Pandas Series are great for one-dimensional data, but in data science, we often work with two-dimensional data tables. This is where Pandas DataFrames come into play. A DataFrame is a two-dimensional labeled data structure with columns of potentially different types. You can think of it as a spreadsheet or SQL table, or a dictionary of Series objects.

2.8.2.2.1 Creating DataFrames

You can create a DataFrame from various data sources, such as dictionaries, lists of lists, or NumPy arrays. Here's an example of creating a DataFrame from a dictionary:

```
data = {
    'Name': ['Alba', 'Jesus', 'Yang'],
    'Age': [30, 25, 35],
    'City': ['New York', 'Los Angeles', 'Chicago']
}
df = pd.DataFrame(data)
df = df.set_index('Name')  # Set 'Name' as the index
print("DataFrame:\n", df)
```

```
DataFrame:
        Age         City
Name
Alba    30     New York
Jesus   25  Los Angeles
Yang    35      Chicago
```

You can also create a DataFrame from a list of lists:

```
# Creating a DataFrame from a list of lists
pd.DataFrame(
```

```
    data=[
        [1, 2, 3],
        [4, 5, 6],
        [7, 8, 9]
    ],
    index=["R1", "R2", "R3"],
    columns=["C1", "C2", "C3"]
)
```

	C1	C2	C3
R1	1	2	3
R2	4	5	6
R3	7	8	9

There are several more ways to create DataFrames, including from CSV files, Excel files, SQL databases, and more. Most of the time, you'll be loading data from external sources rather than creating DataFrames from scratch.

Indexing works similarly to Series, but now you have both row and column labels to consider. Here are some common ways to index and select data in a DataFrame:

Method	Description
`df[column_label]` or `df.column_label` or `df.loc[:, column_label]`	Access a single column by label (returns a Series)
`df[[col1, col2]]`	Access multiple columns by label (returns a DataFrame)
`df.loc[row_labels, column_labels]`	Access rows and columns by **label** (names)
`df.iloc[row_positions, column_positions]`	Access rows and columns by **position** (integers)
`df[boolean_condition]`	Filter rows based on a boolean condition

Consider the following DataFrame

```
df = pd.DataFrame(
    data={
        "area":           ["USA", "Eurozone", "Japan", "UK",
         ↪  "Canada", "Australia"],
        "year":           [2024, 2024, 2024, 2024, 2024, 2024],
        "gdp_growth":     [2.1, 1.3, 0.7, 1.5, 1.8, 2.0],      #
         ↪  in percent
```

```
        "inflation":      [3.2, 2.5, 1.0, 2.8, 2.2, 2.6],      #
         ↪  in percent
        "policy_rate":    [5.25, 4.00, -0.10, 5.00, 4.75, 4.35],
         ↪  # in percent
        "unemployment":   [3.8, 6.5, 2.6, 4.2, 5.1, 4.0],      #
         ↪  in percent
        "fx_usd":         [1.00, 1.09, 143.5, 0.79, 1.36, 1.51] #
         ↪  USD per unit of local currency
    },
    index=["A", "B", "C", "D", "E", "F"]
)
df
```

	area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
A	USA	2024	2.1	3.2	5.25	3.8	1.00
B	Eurozone	2024	1.3	2.5	4.00	6.5	1.09
C	Japan	2024	0.7	1.0	-0.10	2.6	143.50
D	UK	2024	1.5	2.8	5.00	4.2	0.79
E	Canada	2024	1.8	2.2	4.75	5.1	1.36
F	Australia	2024	2.0	2.6	4.35	4.0	1.51

First, we will set the "areas" column as the index of the DataFrame. This will allow us to access rows by area name. We can do this using the `set_index()` method.

```
df = df.set_index("area")
```

We could also do it in-place (modifying the original DataFrame directly)

```
df.set_index("area", inplace=True)
```

2.8.2.2.2 Inspecting DataFrames

You can inspect the first few rows of a DataFrame using the `head()` method and the last few rows using the `tail()` method. By default, both methods display 5 rows, but you can specify a different number as an argument.

```
df.head()  # First 5 rows
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
USA	2024	2.1	3.2	5.25	3.8	1.00
Eurozone	2024	1.3	2.5	4.00	6.5	1.09
Japan	2024	0.7	1.0	-0.10	2.6	143.50

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
UK	2024	1.5	2.8	5.00	4.2	0.79
Canada	2024	1.8	2.2	4.75	5.1	1.36

```
df.tail(3)  # Last 3 rows
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
UK	2024	1.5	2.8	5.00	4.2	0.79
Canada	2024	1.8	2.2	4.75	5.1	1.36
Australia	2024	2.0	2.6	4.35	4.0	1.51

You can get a summary of the DataFrame using the `info()` method, which provides information about the index, columns, data types, and memory usage.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, USA to Australia
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   year          6 non-null      int64
 1   gdp_growth    6 non-null      float64
 2   inflation     6 non-null      float64
 3   policy_rate   6 non-null      float64
 4   unemployment  6 non-null      float64
 5   fx_usd        6 non-null      float64
dtypes: float64(5), int64(1)
memory usage: 336.0+ bytes
```

You can get basic statistical details of the DataFrame using the `describe()` method, which provides measures like mean, standard deviation, min, max, and quartiles for numerical columns.

```
df.describe()
```

	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
count	6.0	6.000000	6.000000	6.000000	6.000000	6.000000
mean	2024.0	1.566667	2.383333	3.875000	4.366667	24.875000
std	0.0	0.520256	0.754763	1.998187	1.318585	58.114711
min	2024.0	0.700000	1.000000	-0.100000	2.600000	0.790000

	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
25%	2024.0	1.350000	2.275000	4.087500	3.850000	1.022500
50%	2024.0	1.650000	2.550000	4.550000	4.100000	1.225000
75%	2024.0	1.950000	2.750000	4.937500	4.875000	1.472500
max	2024.0	2.100000	3.200000	5.250000	6.500000	143.500000

2.8.2.2.3 Indexing and Selecting DataFrames

We can get a single column as a Series using python's getitem syntax on the DataFrame object.

```
df['inflation'] # returns a series
```

```
area
USA          3.2
Eurozone     2.5
Japan        1.0
UK           2.8
Canada       2.2
Australia    2.6
Name: inflation, dtype: float64
```

```
type(df['inflation'])
```

```
pandas.core.series.Series
```

…or using attribute syntax.

```
df.inflation  # returns a series
```

```
area
USA          3.2
Eurozone     2.5
Japan        1.0
UK           2.8
Canada       2.2
Australia    2.6
Name: inflation, dtype: float64
```

If we use a list of column names, we get a DataFrame back

```
df[['inflation']]  # returns a DataFrame
```

area	inflation
USA	3.2
Eurozone	2.5

area	inflation
Japan	1.0
UK	2.8
Canada	2.2
Australia	2.6

```
type(df[['inflation']])
```

```
pandas.core.frame.DataFrame
```

This is useful for selecting multiple columns at once.

```
df[['inflation', 'unemployment']]  # returns a dataframe with
 ↪  selected columns
```

area	inflation	unemployment
USA	3.2	3.8
Eurozone	2.5	6.5
Japan	1.0	2.6
UK	2.8	4.2
Canada	2.2	5.1
Australia	2.6	4.0

We can use `.loc` to select rows and columns by label, and `.iloc` to select rows and columns by position.

- `.loc` uses labels (names) for both rows and columns. The syntax is `df.loc[rows, columns]`. Both can be single labels, lists, or slices. Slices with `.loc` are **inclusive** of the end.
- `.iloc` uses integer positions (like Python lists). The syntax is `df.iloc[rows, columns]`. Slices with `.iloc` are **exclusive** of the end (like standard Python slicing).

Suppose `df` looks like this:

-	name	age	city
0	Alice	23	Madrid
1	Bob	34	London
2	Carol	29	Berlin

- `df['age']` or `df.age` -> Series with ages.

- `df[['name', 'city']]` -> DataFrame with just name and city columns.
- `df.loc[1, 'city']` -> `'London'` (row label 1, column 'city').
- `df.loc[0:1, ['name', 'age']]` -> Rows 0 to 1, columns 'name' and 'age' (inclusive).
- `df.iloc[0:2, 1:3]` -> Rows 0 to 1, columns 1 and 2 (note that row 2 and column 3 are not included).
- `df[df['age'] > 25]` -> Rows where age is greater than 25.

As indicated above, both `.loc` and `.iloc` can take single labels/positions, lists of labels/positions, or slices. Here are some additional tips:

- Use `:` to select all rows or columns:
  - `df.loc[:, 'age']` (all rows, 'age' column).
  - `df.iloc[1, :]` (row 1, all columns).
- Remember: `.loc` is label-based and **inclusive**; `.iloc` is position-based and **exclusive**.

```
df.loc["UK","gdp_growth"] # get the value in row "UK" and column
↪  "gdp_growth"
```

```
np.float64(1.5)
```

```
df.iloc[3,1] # get the value in row 3 and column 1 (recall:
↪  python uses zero-based index)
```

```
np.float64(1.5)
```

You can also get subsets of rows and columns using slices or lists

```
df.loc["USA":"UK",["policy_rate", "fx_usd"]] # Subset rows from
↪  "USA" to "UK" and columns "policy_rate" and "fx_usd"
```

area	policy_rate	fx_usd
USA	5.25	1.00
Eurozone	4.00	1.09
Japan	-0.10	143.50
UK	5.00	0.79

We can filter rows based on a boolean condition.

```
df[df['unemployment'] > 5.0]  # returns a dataframe with rows
↪  where unemployment is greater than 5.0
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
Eurozone	2024	1.3	2.5	4.00	6.5	1.09
Canada	2024	1.8	2.2	4.75	5.1	1.36

To filter rows in a DataFrame based on multiple conditions, you can use logical operators:

Operator	Symbol	Meaning	General Pattern
AND	`&`	All conditions must be true	`df[(condition1) & (condition2)]`
OR	`|`	At least one condition must be true	`df[(condition1) | (condition2)]`
NOT	`~`	Negates a condition (condition is false)	`df[~(condition)]`

You can combine these operators to build more complex filters as needed. For example

```
df[(condition1 & condition2) | (~condition3 & condition4)]
```

To reduce the likelihood of mistakes, always enclose each condition in parentheses to ensure correct evaluation.

The following example filters the DataFrame to include only rows where the `fx_usd` is less than 1.0 **and** the `inflation` is greater than 2.0:

```
df[(df['fx_usd'] < 1.0) & (df["inflation"] > 2.0)]
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd
UK	2024	1.5	2.8	5.0	4.2	0.79

2.8.2.2.4 DataFrame Operations

There are many operations you can perform on DataFrames. Here are some common ones:

Adding Columns:

Method	Code Pattern (Abstraction)	Notes
Direct assign	`df[new_col] = values`	Adds or overwrites a column
`assign()`	`df.assign(new_col=values)`	Adds a new column (returns a new DataFrame)
`insert()`	`df.insert(loc, new_col, values)`	Adds at specific position
Multiple cols	`df[[col1, col2]] = values`	Assign multiple columns at once

Adding Rows:

Method	Code Pattern (Abstraction)	Notes
`loc`	`df.loc[new_label] = values`	Adds or overwrites a row by index label
`iloc`	`df.iloc[position] = values`	Overwrites a row at a specific integer position (does not add a new row)
`concat()`	`df = pd.concat([df, new_rows_df])`	Adds one or more new rows from another DataFrame

For example, to add a new column that approximates real GDP growth (i.e., nominal GDP growth minus inflation):

```
df["real_gdp_growth"] = df.gdp_growth - df.inflation  # Create a
 ↪  new column as the difference between gdp_growth and inflation
df["avg_weather"] = [20.5, 18.0, 15.0, 12.5, 10.0, 22.0]  # Add a
 ↪  new column with average weather data
df
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd	real_gdp_growt
USA	2024	2.1	3.2	5.25	3.8	1.00	-1.1
Eurozone	2024	1.3	2.5	4.00	6.5	1.09	-1.2
Japan	2024	0.7	1.0	-0.10	2.6	143.50	-0.3
UK	2024	1.5	2.8	5.00	4.2	0.79	-1.3
Canada	2024	1.8	2.2	4.75	5.1	1.36	-0.4
Australia	2024	2.0	2.6	4.35	4.0	1.51	-0.6

Using `assign()`, we can do the same without modifying the original DataFrame (note that `assign()` returns a new DataFrame):

```
df = df.drop(columns=["real_gdp_growth"])  # Remove previously
 ↪  added column
df_new = df.assign(real_gdp_growth=df.gdp_growth - df.inflation)
df_new
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weather	real_gdp_gro
USA	2024	2.1	3.2	5.25	3.8	1.00	20.5	-1.1
Eurozone	2024	1.3	2.5	4.00	6.5	1.09	18.0	-1.2
Japan	2024	0.7	1.0	-0.10	2.6	143.50	15.0	-0.3
UK	2024	1.5	2.8	5.00	4.2	0.79	12.5	-1.3
Canada	2024	1.8	2.2	4.75	5.1	1.36	10.0	-0.4
Australia	2024	2.0	2.6	4.35	4.0	1.51	22.0	-0.6

Using `insert()`, we can add a new column at a specific position. For example, to insert a `gdp_per_capita` column as the second column (index 1):

```
df.insert(
    loc=1,  # Insert at the second position (0-based index)
    column='gdp_per_capita',  # Name of the new column
    value=[60000, np.nan, 40000, np.nan, 55000, 70000]  # Values
 ↪  for the new column
)
df
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0

Deleting data:

What to Remove	Method/Option	Code Pattern (Abstraction)	Notes
Columns by label	`drop()`	`df.drop([col1, col2, ...], axis=1)`	Returns new DataFrame

What to Remove	Method/Option	Code Pattern (Abstraction)	Notes
Columns by label (in-place)	`drop()`	`df.drop([col1, col2, ...], axis=1, inplace=True)`	Modifies original DataFrame
Columns by position	`drop()`	`df.drop(df.columns[[pos1, pos2, ...]], axis=1)`	Use integer positions
Columns with missing values	`dropna()`	`df.dropna(axis=1)`	Removes columns with any missing
Rows by label	`drop()`	`df.drop([row1, row2, ...], axis=0)`	Returns new DataFrame
Rows by label (in-place)	`drop()`	`df.drop([row1, row2, ...], axis=0, inplace=True)`	Modifies original DataFrame
Rows by position	`drop()`	`df.drop(df.index[[pos1, pos2, ...]], axis=0)`	Use integer positions
Rows with missing values	`dropna()`	`df.dropna(axis=0)`	Removes rows with any missing
Duplicate rows	`drop_duplicates()`	`df.drop_duplicates()`	Removes duplicate rows

For example, to remove the `avg_weather` column we just added

```
df.drop("avg_weather", axis=1)
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51

We can also drop columns with NaN values

```
df.dropna(axis=1)  # Drops columns with any NaN values
```

area	year	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weather
USA	2024	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	2.0	2.6	4.35	4.0	1.51	22.0

Or fill it up with default "fallback" data:

```
df.fillna(df.gdp_per_capita.median())  # Fills NaN values with
 ↪  the median of the gdp_per_capita column
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	57500.0	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	57500.0	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0

Note that both `drop()` and `fillna()` return a new DataFrame by default. Thus, when we access `df` again, we will see that it still contains the `avg_weather` column and any NaN values.

```
df  # Original DataFrame remains unchanged
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0

We can also sort the entries in dataframes, e.g. alphabetically by index or numerically by column values

What to Sort	Method/Option	Code Pattern (Abstraction)	Notes
By column(s)	`sort_values()`	`df.sort_values(by=col)`	Sort by one column (ascending by default)
By multiple columns	`sort_values()`	`df.sort_values(by=[col1, col2])`	Sort by several columns (priority order)
By column(s), descending	`sort_values()`	`df.sort_values(by=col, ascending=False)`	Sort in descending order
By multiple columns, custom order	`sort_values()`	`df.sort_values(by=[col1, col2], ascending=[True, False])`	Custom order for each column
By index	`sort_index()`	`df.sort_index()`	Sort by row index (ascending by default)
By index, descending	`sort_index()`	`df.sort_index(ascending=False)`	Sort index in descending order
By columns (column labels)	`sort_index()`	`df.sort_index(axis=1)`	Sort columns by their labels
By columns, descending	`sort_index()`	`df.sort_index(axis=1, ascending=False)`	Sort columns in descending order

For example, to sort the DataFrame by `inflation` in descending order

```
df.sort_values(by='inflation', ascending=False)
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50

To sort by multiple columns, e.g., first by `year` (ascending) and then by `gdp_growth` (descending):

```
df.sort_values(by=['year', 'gdp_growth'], ascending=[True,
 ↪  False])
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0

We can also sort by index

```
df.sort_index()
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5

or column names

```
df.sort_index(axis=1)
```

area	avg_weather	fx_usd	gdp_growth	gdp_per_capita	inflation	policy_rate	unemployment	y
USA	20.5	1.00	2.1	60000.0	3.2	5.25	3.8	2
Eurozone	18.0	1.09	1.3	NaN	2.5	4.00	6.5	2
Japan	15.0	143.50	0.7	40000.0	1.0	-0.10	2.6	2
UK	12.5	0.79	1.5	NaN	2.8	5.00	4.2	2
Canada	10.0	1.36	1.8	55000.0	2.2	4.75	5.1	2
Australia	22.0	1.51	2.0	70000.0	2.6	4.35	4.0	2

Pandas supports a wide range of methods for merging different datasets. These are described extensively in the documentation. Here we just give a few examples.

Method	Function	Description	Key Parameters	Use Case
Inner Join	`pd.merge(df1, df2, how='inner')`	Returns only rows with matching keys in both dataframes	`on`, `left_on`, `right_on`	When you only want records that exist in both datasets
Left Join	`pd.merge(df1, df2, how='left')`	Returns all rows from left dataframe, matching rows from right	`on`, `left_on`, `right_on`	Keep all records from primary dataset, add matching info
Right Join	`pd.merge(df1, df2, how='right')`	Returns all rows from right dataframe, matching rows from left	`on`, `left_on`, `right_on`	Keep all records from secondary dataset
Outer Join	`pd.merge(df1, df2, how='outer')`	Returns all rows from both dataframes	`on`, `left_on`, `right_on`	When you want all records from both datasets
Cross Join	`pd.merge(df1, df2, how='cross')`	Cartesian product of both dataframes	None required	Create all possible combinations
Concat Vertical	`pd.concat([df1, df2])`	Stacks dataframes vertically (rows)	`axis=0`, `ignore_index`	Combine datasets with same columns
Concat Horizontal	`pd.concat([df1, df2], axis=1)`	Joins dataframes horizontally (columns)	`axis=1`, `join`	Combine datasets with same index
Join Method	`df1.join(df2)`	Left join based on index	`how`, `lsuffix`, `rsuffix`	Quick join on index when columns don't overlap

```
df_trade = pd.DataFrame({
    "area": ["USA", "Eurozone", "Japan", "China", "India",
     ↪  "Brazil"],
    "exports_bn": [1650, 2200, 705, 3360, 323, 281],
    "imports_bn": [2407, 2000, 641, 2601, 507, 219],
    "trade_balance": [-757, 200, 64, 759, -184, 62]
}).set_index("area")
df_trade
```

area	exports_bn	imports_bn	trade_balance
USA	1650	2407	-757
Eurozone	2200	2000	200
Japan	705	641	64
China	3360	2601	759
India	323	507	-184
Brazil	281	219	62

```
inner_result = pd.merge(df, df_trade, how='inner',
 ↪  left_index=True, right_index=True)
inner_result
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0

```
left_result = pd.merge(df, df_trade, how='left', left_index=True,
 ↪  right_index=True)
left_result
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_weat
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0

```
right_result = pd.merge(df, df_trade, how='right',
 ↪  left_index=True, right_index=True)
right_result
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_us
USA	2024.0	60000.0	2.1	3.2	5.25	3.8	1.00
Eurozone	2024.0	NaN	1.3	2.5	4.00	6.5	1.09
Japan	2024.0	40000.0	0.7	1.0	-0.10	2.6	143.5
China	NaN	NaN	NaN	NaN	NaN	NaN	NaN
India	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Brazil	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
outer_result = pd.merge(df, df_trade, how='outer',
 ↪  left_index=True, right_index=True)
outer_result
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_us
Australia	2024.0	70000.0	2.0	2.6	4.35	4.0	1.51
Brazil	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Canada	2024.0	55000.0	1.8	2.2	4.75	5.1	1.36
China	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Eurozone	2024.0	NaN	1.3	2.5	4.00	6.5	1.09
India	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Japan	2024.0	40000.0	0.7	1.0	-0.10	2.6	143.5
UK	2024.0	NaN	1.5	2.8	5.00	4.2	0.79
USA	2024.0	60000.0	2.1	3.2	5.25	3.8	1.00

```
pd.concat([df, df_trade], axis=1).sort_index()  # Concatenate
 ↪  along columns
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_us
Australia	2024.0	70000.0	2.0	2.6	4.35	4.0	1.51
Brazil	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Canada	2024.0	55000.0	1.8	2.2	4.75	5.1	1.36
China	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Eurozone	2024.0	NaN	1.3	2.5	4.00	6.5	1.09
India	NaN	NaN	NaN	NaN	NaN	NaN	NaN

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_we
Japan	2024.0	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024.0	NaN	1.5	2.8	5.00	4.2	0.79	12.5
USA	2024.0	60000.0	2.1	3.2	5.25	3.8	1.00	20.5

```
pd.concat([df, df_trade], axis=0)  # Concatenate along rows
```

area	year	gdp_per_capita	gdp_growth	inflation	policy_rate	unemployment	fx_usd	avg_we
USA	2024.0	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024.0	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024.0	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024.0	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024.0	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024.0	70000.0	2.0	2.6	4.35	4.0	1.51	22.0
USA	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Eurozone	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Japan	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
China	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
India	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Brazil	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Sometimes it can be useful to apply a function to all values of a column/row. For instance, we might be interested in normalised inflation. We can do this using the `apply()` method. This method applies a function to each element in the Series or DataFrame.

```
df.inflation.apply(lambda x: (x - df.inflation.mean()) /
 ↪  df.inflation.std())  # Standardize the inflation column
```

```
area
USA          1.082018
Eurozone     0.154574
Japan       -1.832806
UK           0.552050
Canada      -0.242902
Australia    0.287066
Name: inflation, dtype: float64
```

Sometimes it is necessary to rename columns or indices in a DataFrame. There are several ways to do this, depending on whether you want to rename all columns, specific columns, or apply a function to transform the names.

Method	Syntax	Use Case	Example
Direct Assignment	`df.columns = [list]`	Replace all column names at once	`df.columns = ['A', 'B', 'C']`
rename() with Dictionary	`df.rename(columns={dict})`	Rename specific columns selectively	`df.rename(columns={'old': 'new'})`
rename() with inplace	`df.rename(columns={dict}, inplace=True)`	Modify original DataFrame directly	`df.rename(columns={'old': 'new'}, inplace=True)`
rename() with Function	`df.rename(columns=function)`	Apply transformation to all columns	`df.rename(columns=str.upper)`
String Methods	`df.columns.str.method()`	Apply string operations to column names	`df.columns = df.columns.str.replace('_', ' ')`
Lambda Function	`df.rename(columns=lambda x: expression)`	Custom transformations on column names	`df.rename(columns=lambda x: x.replace('old', 'new'))`

Key Parameters

Parameter	Description	Default	Example
`columns`	Dictionary or function for column mapping	`None`	`{'old_name': 'new_name'}`
`inplace`	Modify DataFrame in place vs. return copy	`False`	`inplace=True`
`errors`	How to handle missing keys	`'ignore'`	`errors='raise'`

```
df1 = df.copy()  # Create a copy of the DataFrame

df1 = df1.rename(columns={
    "gdp_growth": "gdp_growth_(%)",
    "gdp_per_capita": "gdp_per_capita_($)",
    "inflation": "inflation_rate_(%)",
    "policy_rate": "policy_rate_(%)",
    "unemployment": "unemployment_rate_(%)",
    "fx_usd": "fx_rate_($/X)",
    "avg_weather": "avg_weather_(°C)",
    })  # Rename columns
df1
```

area	year	gdp_per_capita_($)	gdp_growth_(%)	inflation_rate_(%)	policy_rate_(%)	unemploym
USA	2024	60000.0	2.1	3.2	5.25	3.8
Eurozone	2024	NaN	1.3	2.5	4.00	6.5
Japan	2024	40000.0	0.7	1.0	-0.10	2.6
UK	2024	NaN	1.5	2.8	5.00	4.2
Canada	2024	55000.0	1.8	2.2	4.75	5.1
Australia	2024	70000.0	2.0	2.6	4.35	4.0

We can also work directly with column names

```
df1.columns = df.columns.str.replace('_', ' ')
df1
```

area	year	gdp per capita	gdp growth	inflation	policy rate	unemployment	fx usd	avg weather
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20.5
Eurozone	2024	NaN	1.3	2.5	4.00	6.5	1.09	18.0
Japan	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15.0
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12.5
Canada	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10.0
Australia	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22.0

or the row names

```
# Capitalize the first letter of each area name
df1.index = df.index.str.upper() # Convert all area names to
 ↪  uppercase
df1.columns = df.columns.str.capitalize()  # Capitalize the first
 ↪  letter of each column name
df1
```

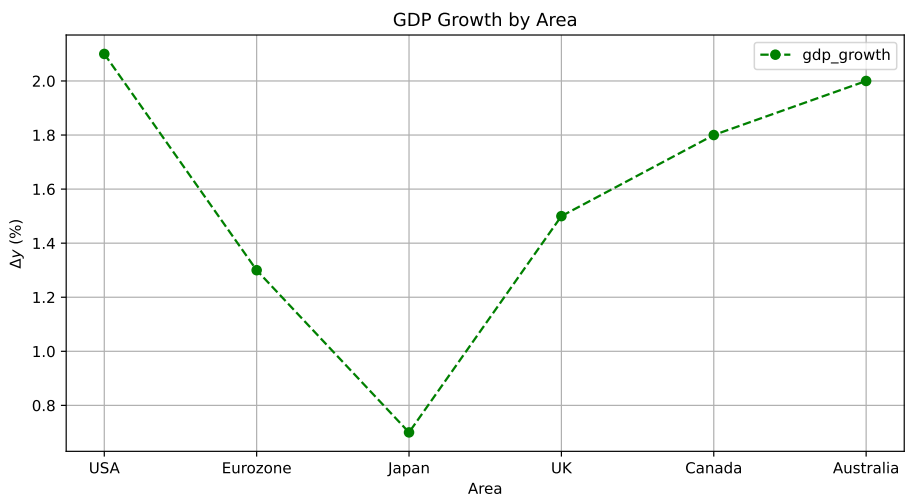
area	Year	Gdp_per_capita	Gdp_growth	Inflation	Policy_rate	Unemployment	Fx_usd	Av
USA	2024	60000.0	2.1	3.2	5.25	3.8	1.00	20
EUROZONE	2024	NaN	1.3	2.5	4.00	6.5	1.09	18
JAPAN	2024	40000.0	0.7	1.0	-0.10	2.6	143.50	15
UK	2024	NaN	1.5	2.8	5.00	4.2	0.79	12
CANADA	2024	55000.0	1.8	2.2	4.75	5.1	1.36	10
AUSTRALIA	2024	70000.0	2.0	2.6	4.35	4.0	1.51	22

2.8.2.3 Data Visualization with Pandas

DataFrames have all kinds of useful plotting built in. Note that by default `matplotlib` is used as the backend for plotting in Pandas. However, Pandas imports matplotlib for you in the background and you don't have to do it yourself.

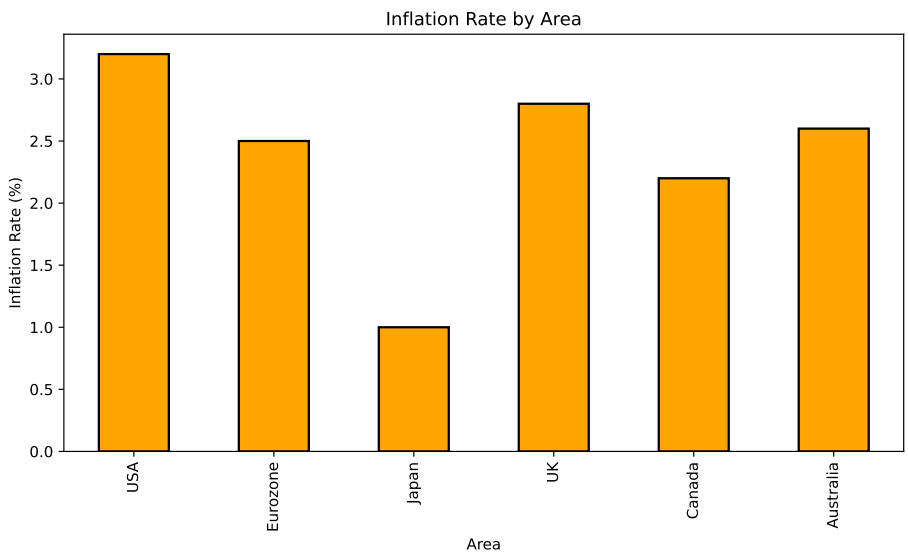
You can create various types of plots directly from DataFrames and Series using the `plot()` method. Here are some examples:

```
df.gdp_growth.plot(
  kind='line',
  title='GDP Growth by Area',
  ylabel=r'$ \Delta y$ (%)',
  xlabel='Area',
  grid=True,
  figsize=(10, 5),
  legend=True,
  color='green',
  marker='o',
  linestyle='--'
)
```

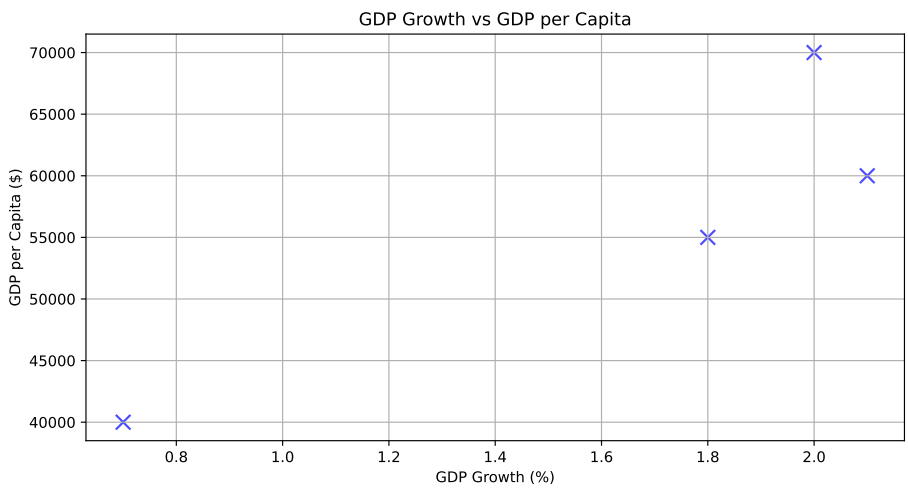



```
df.inflation.plot(
  kind='bar',
  title='Inflation Rate by Area',
  ylabel='Inflation Rate (%)',
  xlabel='Area',
  color="orange",
  grid=False,
  figsize=(10, 5),
  legend=False,
```

```
  edgecolor='black',
  linewidth=1.5
)
```




```
df.plot(
  kind="scatter",
  x="gdp_growth",
  y="gdp_per_capita",
  title="GDP Growth vs GDP per Capita",
  xlabel="GDP Growth (%)",
  ylabel="GDP per Capita ($)",
  grid=True,
  figsize=(10, 5),
  color="blue",
  marker="x",
  s=100,  # Size of the markers
  alpha=0.7,  # Transparency of the markers
  linewidth=1.5 # Edge width of the markers
)
```



2.8.2.4 Importing and Exporting Data

We have seen how to create DataFrames from scratch. However, in practice, we often need to load data from external files or databases. Pandas provides a variety of functions to read and write data in different formats. Data can be imported from CSV, Excel, and more. To read a CSV file into a DataFrame, you can use the `pd.read_csv()` function.

```
file_csv ='./data.csv'
data = pandas.read_csv(file_csv)
```

To read an Excel file, you can use the `pd.read_excel()` function.

```
file_excel = './data.xlsx'
data = pandas.read_excel(file_excel, sheet_name='Sheet1')
```

To write a DataFrame to a CSV file, you can use the `to_csv()` method.

```
df.to_csv('output.csv', index=False)
```

To write a DataFrame to an Excel file, you can use the `to_excel()` method.

```
df.to_excel('output.xlsx', sheet_name='Sheet1', index=False)
```

We will cover these and other data I/O methods in more detail in later sections of the course.

2.8.3 Visualization: Matplotlib & Seaborn

The library that will allow us to do data visualization, whether static, animated or interactive, in Python is Matplotlib.

Matplotlib plots our data on Figures, each one containing axes where data points are defined as x-y coordinates. This means that matplotlib will allow us to manipulate the displays by changing the elements of these two classes:

Figure:

The entire figure that contains the axes, which are the actual plots.

Axes:

Contains a region for plotting data, and include the x and y axis (and z if is 3D) where we actually plot the data.

> **Note**
>
> Documentations for these packages are available at https://matplotlib.org/stable/ and https://seaborn.pydata.org/api.html.

We can import Matplotlib as follows:

```
import matplotlib.pyplot as plt
```

2.9 Working with Application Programming Interfaces (APIs)

Application Programming Interfaces (APIs) are a set of rules and protocols that allow different software applications to communicate with each other. They enable developers to access data and functionality from external services, libraries, or platforms without needing to understand the underlying code or infrastructure. Rather than downloading data files manually, APIs allow us to programmatically request and retrieve data directly from a web service.

In this section, we will have a brief look at how to use some common APIs for economic data retrieval using Python. We will cover the following:

- Banco de España's Statistics Web Service
- ECB Data Portal
- Fred API by the Federal Reserve Bank of St. Louis

These APIs provide access to a wide range of economic and financial data, including interest rates, exchange rates, inflation rates, GDP figures, and more. By using these APIs, we can automate the process of data retrieval, ensuring that we always have access to the most up-to-date information for our analyses. I highly recommend that you make use of APIs whenever possible to streamline your data collection process.

2.9.1 Banco de España's Statistics Web Service

Banco de España's Statistics Web Service provides a way to programmatically retrieve data from the Banco de España's databases including data from BIEST. Since Banco de España does not provide an official Python package to access their API, we can use the `requests` library to make HTTP requests and retrieve data in JSON (JavaScript Object Notation) format. We can then parse the JSON data and convert it into a Pandas DataFrame for further analysis.

To this end, we first import the necessary libraries

```
import requests
import pandas as pd
```

Next, we define a class to interact with the Banco de España API[1]

```
class BancoDeEspanaAPI:
    def __init__(self, language='en'):
      self.language = language

    def request(self, url):
      response = requests.get(url)
      return response.json()

    def get_series(self, series, time_range='MAX'):

      # Prepare the series parameter
      if isinstance(series, list):
          series_list = ','.join(series)
      else:
          series_list = series

      # Download the data for the specified series
      url = f"https://app.bde.es/bierest/resources/srdatosapp/li
↪ staSeries?idioma={self.language}&series={series_list}&rango=
↪ {time_range}"
      json_response = self.request(url)

      # Initialize an empty dataframe to store the results
      df = pd.DataFrame()

      # Go over each series in the response and extract the data
      for series in json_response:

        # Extract series name, dates, and values
        series_name = series['serie']
```

[1]Note that creating the class is not strictly necessary, but it helps to organize the code.

```
        dates = series['fechas']
        values = series['valores']

        # Add the data to the dataframe
        df[series_name] = pd.Series(data=values,
 ↪  index=pd.to_datetime(dates).date)

        # Sort the dataframe by index (date)
        df = df.sort_index()

      return df
```

We can then create an instance of the `BancoDeEspanaAPI` class and use its methods to retrieve data. For example, to get the latest data for a specific series, we can use the `latest_data()` method

```
bde = BancoDeEspanaAPI()
df = bde.get_series(['DTNPDE2010_P0000P_PS_APU',
 ↪  'DTNSEC2010_S0000P_APU_SUMAMOVIL'])
```

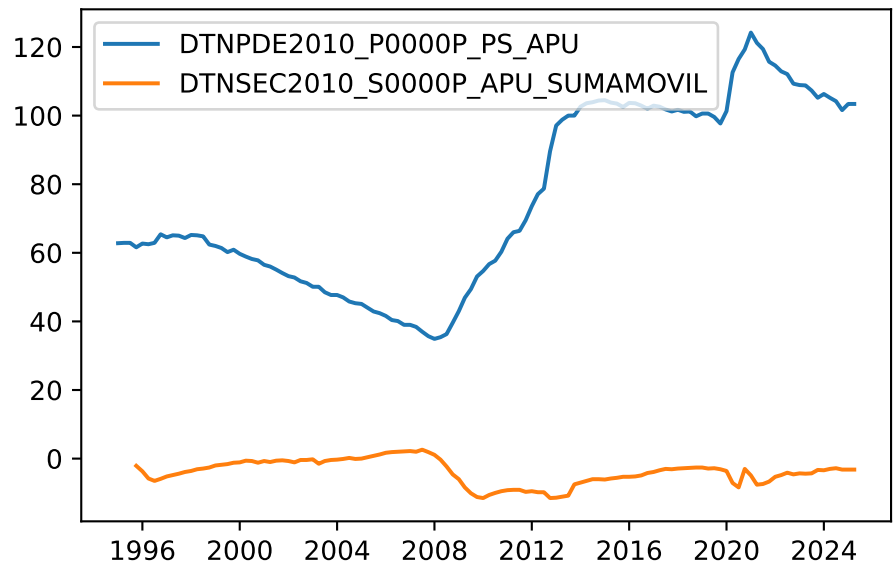
Now, the requested series are in the DataFrame `df` and we can manipulate or analyze them as needed. For example, we can display the retrieved data

```
df.tail()
```

	DTNPDE2010_P0000P_PS_APU	DTNSEC2010_S0000P_APU_SUMAMOVIL
2024-04-01	105.2	-3.0
2024-07-01	104.2	-2.8
2024-10-01	101.6	-3.2
2025-01-01	103.4	-3.2
2025-04-01	103.4	-3.2

or plot it

```
df.plot()
```



This is a very basic implementation of how to interact with the Banco de España API using Python. You can extend this class to include more functionality, such as handling different data formats, error handling, and more advanced data processing as needed. To get the series keys for the data you want to retrieve, you can use the BIEST tool provided by Banco de España.

2.9.2 ECB Data Portal & Other SDMX APIs

The ECB Data Portal provides access to a wide range of economic and financial data from the European Central Bank. Similar to Banco de España, the ECB does not provide an official Python package for their API. However, the ECB follows the SDMX standard for data exchange, which allows us to retrieve data in a structured format. We can use the `sdmx` library in Python to interact with the ECB API and retrieve data.

First, we need to import the necessary libraries

```
import sdmx
import pandas as pd
```

Then, we need to initialize a connection to the ECB API

```
ecb = sdmx.Client("ECB")
```

Suppose we want to retrieve the HICP inflation rate for Spain from January 2019 to June 2019. This series has the following key: `ICP.M.ES.N.000000.4.ANR`.

To download it we need to specify the appropriate parameters and make a request

to the ECB API

```
key = 'M.ES.N.000000.4.ANR' # Need key without the 'ICP.' prefix
params = dict(startPeriod="2019-01", endPeriod="2019-06") # This
 ↪  is optional
data = ecb.data("ICP", key=key, params=params).data[0] # ICP
 ↪  prefix needs to be specified here
df = sdmx.to_pandas(data).to_frame()
```

Now, the requested data is in the DataFrame `df` and we can manipulate or analyze it as needed. For example, we can display the retrieved data

```
df.tail()
```

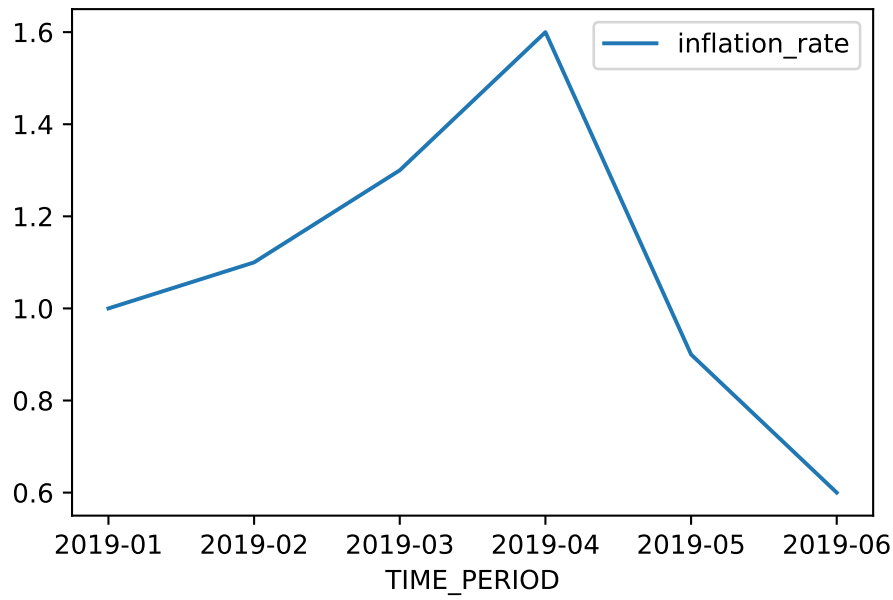
FREQ	REF_AREA
M	ES

Note that this is a multi-index DataFrame. We can reset the index to make it easier to work with

```
df = df.reset_index()
df = df.set_index('TIME_PERIOD')
df = df.loc[:, ['value']]
df = df.rename(columns={'value': 'inflation_rate'})
```

We can plot the data as usual

```
df.plot()
```



These are just basic examples of how to interact with the ECB API using Python. The `sdmx` library supports many more features.

> **Other SDMX Data Providers**
>
> The SDMX standard is used by various international organizations for data exchange. Some other notable SDMX APIs include:
> - Eurostat
> - Bank for International Settlements (BIS)
> - International Monetary Fund (IMF)
> - OECD
>
> You can find a list of SDMX data provides implemented in the `sdmx` package here. To use them in the code above you simply need to replace `'ECB'` with the appropriate provider name.

2.9.3 Fred API

The Fred API by the Federal Reserve Bank of St. Louis provides access to a vast amount of economic data, including interest rates, inflation rates, GDP figures, and more. To use the Fred API, we need to sign up for an API key on the Fred website. Once we have the API key, we can use the `pyfredapi` library in Python to interact with the Fred API and retrieve data.

The Fred API works a little differently from the previous two APIs we have seen since it requires an API key for authentication. You can sign up for a free API

key on the Fred website. Note that these keys are personal and should not be shared publicly. For this reason, the key is not included directly in the code examples below. Instead, you should follow the instructions in the `pyfredapi` documentation to set up your API key securely.

Once we have set the API key, we need to import the necessary libraries

```
import pyfredapi as pf
```

Then, we can download the series for GDP (series ID: `GDP`) as follows

```
df = pf.get_series('GDP') # Note that you can provide the API key
 ↪  manually by adding the parameter api_key='YOUR_API_KEY' if
 ↪  you have not set it up as an environment variable
```

We can then display the retrieved data

```
df.tail()
```

	realtime_start	realtime_end	date	value
313	2025-09-25	2025-09-25	2024-04-01	29147.044
314	2025-09-25	2025-09-25	2024-07-01	29511.664
315	2025-09-25	2025-09-25	2024-10-01	29825.182
316	2025-09-25	2025-09-25	2025-01-01	30042.113
317	2025-09-25	2025-09-25	2025-04-01	30485.729

Cleaning up the DataFrame a bit

```
df = df.rename(columns={'value': 'gdp'}) # Rename the 'value'
 ↪  column to 'gdp'
df['date'] = pd.to_datetime(df['date']) # Convert the 'date'
 ↪  column to datetime format
df = df.set_index('date') # Set the 'date' column as the index
df = df.loc[:, ['gdp']] # Keep only the 'gdp' column
```

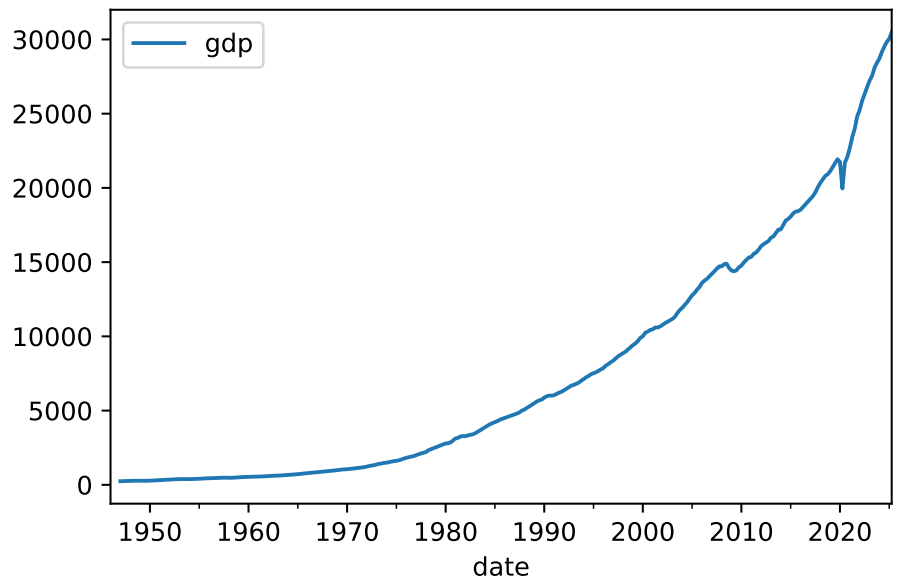
Now it looks better

```
df.tail()
```

	gdp
date	
2024-04-01	29147.044
2024-07-01	29511.664
2024-10-01	29825.182
2025-01-01	30042.113
2025-04-01	30485.729

and to plot it, we can simply do

```
df.plot()
```



To see all the functionality provided by the `pyfredapi` library, please refer to the official documentation.

2.10 Good Practices for Python Programming

As you develop your Python programming skills, adopting good practices early will save you countless hours of frustration and make your code more maintainable, reproducible, and professional. This section covers essential practices that every Python programmer should follow, with particular emphasis on version control and virtual environments—two foundational tools that are often overlooked by beginners but are indispensable in professional settings. Due to their importance, we will briefly cover them here. However, we do not have the time to go into great detail in this course. Therefore, I encourage you to explore these topics further on your own.

2.10.1 Version Control with Git

Version control is perhaps the single most important practice for any programmer. It allows you to track changes to your code over time, collaborate with others, recover from mistakes, and maintain a complete history of your project's evolution. Git is the dominant version control system used in both academia and industry, and GitHub is the most popular platform for hosting Git repositories.

Think of Git as a sophisticated "undo" system for your entire project. Every time you make a commit, you create a snapshot of your project that you can return to at any time. This means you can experiment fearlessly—if your new approach doesn't work, you can simply revert to a previous state. Beyond this safety net, Git enables powerful collaboration workflows: multiple people can work on the same codebase simultaneously, with Git helping to merge their changes intelligently.

For academic research and data science projects, version control is equally crucial. It provides a complete audit trail of your analysis, which is essential for reproducibility. When reviewers ask about a result from six months ago, you can check out the exact code that produced it. When you discover an error, you can trace back to when it was introduced. This level of rigor is increasingly expected in computational research.

To get started with Git for your Python projects, you'll want to follow a basic workflow. First, initialize a Git repository in your project folder using `git init`. As you work, periodically stage your changes with `git add` and commit them with meaningful messages using `git commit -m "Description of changes"`. Push your commits to a remote repository on GitHub to back up your work and enable collaboration. A typical Git workflow looks like this:

```
# Initialize a new Git repository
git init

# Add files to staging area
git add script.py

# Commit changes with a descriptive message
git commit -m "Add data preprocessing function"

# Push to GitHub (after setting up remote)
git push origin main
```

Some best practices for using Git include committing frequently with small, logical changes rather than massive commits that touch many files; writing clear commit messages that explain *why* you made the change, not just *what* changed; and using a `.gitignore` file to exclude data files, output files, and environment-specific files from version control. You should version control your code and configuration files, but avoid committing large datasets, model weights, or generated outputs—these should be stored separately or regenerated from your code.

Many cloud platforms like GitHub offer additional features beyond basic version control. Issues help track bugs and feature requests, pull requests facilitate code review before merging changes, and GitHub Actions can automate testing and deployment.

2.10.2 Virtual Environments and Package Management

Virtual environments are isolated Python installations that allow you to maintain different sets of packages for different projects. This solves a critical problem: different projects often require different versions of the same library. Without virtual environments, you'd be forced to use a single global installation of each package, which can lead to version conflicts and "it works on my machine" problems.

Consider a practical scenario: you're working on an older data analysis project that requires NumPy 1.20, but a new machine learning project needs NumPy 1.24 for compatibility with the latest PyTorch. Without virtual environments, you'd have to constantly uninstall and reinstall NumPy depending on which project you're working on. Virtual environments solve this elegantly by creating separate Python installations for each project, each with its own package versions.

Beyond avoiding conflicts, virtual environments make your projects reproducible. When you share your code with others or run it on a different machine, you need a way to specify exactly which package versions it requires. By creating an environment file (like `environment.yml` for conda or `requirements.txt` for pip), you provide a recipe that others can use to recreate your exact setup. This is essential for reproducible research and collaborative projects.

There several different tools for managing virtual environments in Python. Two commonly used ones are conda and venv. Conda, which comes with Anaconda and Miniconda, is particularly popular in data science because it can manage both Python packages and system-level dependencies. It's especially useful when you need packages that require compiled code, like NumPy or PyTorch. The built-in venv module creates lighter-weight environments but only manages Python packages, requiring you to handle system dependencies separately.

In this course, we used conda to manage virtual environments. To create a virtual environment for a project, you would use:

```
# Create a new environment named 'myproject' with Python 3.11
conda create -n myproject python=3.11

# Activate the environment
conda activate myproject

# Install packages in the activated environment
conda install numpy pandas matplotlib

# Export environment to a file for reproducibility
conda env export > environment.yml

# Create environment from file on another machine
conda env create -f environment.yml
```

Once you've activated an environment, any packages you install or Python scripts you run will use that environment's isolated installation. When you're done working, you can deactivate it with `conda deactivate`. This workflow keeps each project's dependencies cleanly separated.

A good practice is to create a fresh virtual environment at the start of each new project and document its dependencies in an `environment.yml` file. Keep this file in your Git repository so others can recreate your setup. Update the file whenever you add new packages to your project. When sharing your code, include instructions for setting up the environment—this is often just a single command: `conda env create -f environment.yml`.

The combination of Git and virtual environments forms a foundation for reproducible computational work. Git tracks your code changes, while virtual environments ensure your code runs consistently across different machines and over time. Together, they transform ad-hoc scripts into professional, maintainable projects that you and others can build upon.

2.10.3 Code Organization and Documentation

Well-organized and documented code is easier to understand, maintain, and debug. As your projects grow beyond simple scripts, good organization becomes essential. Break your code into logical functions and modules rather than writing everything in a single long script. Each function should do one thing well and have a clear, descriptive name. Use docstrings to document what each function does, what parameters it expects, and what it returns.

Python docstrings are enclosed in triple quotes and appear immediately after a function definition. A good docstring explains the purpose of the function, describes parameters and return values, and may include usage examples. Here's a well-documented function:

```python
def calculate_portfolio_return(weights, returns):
    """
    Calculate the expected return of a portfolio.

    Parameters
    ----------
    weights : array-like
        Portfolio weights for each asset (should sum to 1)
    returns : array-like
        Expected returns for each asset

    Returns
    -------
    float
        Expected portfolio return
```

```
    Examples
    --------
    >>> weights = np.array([0.6, 0.4])
    >>> returns = np.array([0.10, 0.15])
    >>> calculate_portfolio_return(weights, returns)
    0.12
    """
    return np.dot(weights, returns)
```

For larger projects, organize your code into modules (separate `.py` files) grouped by functionality. Use meaningful file and variable names—`data_preprocessing.py` is much clearer than `utils.py`, and `interest_rate` is better than `x`. Follow the PEP 8 style guide for Python code, which covers naming conventions, indentation, and other formatting guidelines.

2.10.4 Error Handling and Debugging

Errors are an inevitable part of programming. Learning to handle them gracefully and debug effectively will make you a much more productive programmer. Python uses exceptions to signal errors. Rather than letting your program crash, you can catch exceptions and handle them appropriately using try-except blocks:

```
try:
    df = pd.read_csv('data.csv')
except FileNotFoundError:
    print("Error: data.csv not found. Please check the file
     ↪  path.")
    df = None
```

When debugging, you can use print statements strategically to understand what's happening in your code, use Python's built-in debugger (pdb) or VSCode's debugging features for more complex issues. The VSCode debugger lets you set breakpoints, step through code line by line, and inspect variable values—invaluable for tracking down subtle bugs.

Don't be discouraged when you encounter errors. Reading error messages carefully is a crucial skill—Python's error messages usually tell you exactly what went wrong and where. The traceback shows the sequence of function calls that led to the error, with the actual error at the bottom. Learning to parse these messages will help you fix issues quickly.

2.10.5 Using AI Tools for Coding

Modern AI tools like GitHub Copilot, and Claude Code can significantly accelerate your coding, especially when you're learning. These tools can help you write boilerplate code, explain unfamiliar syntax, suggest solutions to common

problems, and even debug errors. However, use them thoughtfully—treat them as helpful assistants, not replacements for understanding.

When using AI coding assistants, always read and understand the suggested code before using it. Don't blindly copy-paste without comprehension. These tools can make mistakes or suggest suboptimal solutions, so critical evaluation is essential. Use them to learn: if an AI suggests an unfamiliar approach, research why it works and when it's appropriate. Over time, you'll develop intuition for when AI suggestions are helpful versus when you need to think more carefully about the problem.

AI tools are particularly useful for learning new libraries or APIs, generating test cases, refactoring code, and getting past "blank page" syndrome when starting a new function. They're less reliable for complex algorithmic problems or domain-specific logic that requires deep understanding. Like any tool, they become more valuable as you learn to use them effectively.

Part II

Supervised Learning

Chapter 3

Overview of Supervised Learning

This is a dummy section.

Part III

Natural Language Processing

Chapter 4

Overview of Natural Language Processing (NLP)

This is a dummy section.

Part IV

Generative AI

Chapter 5

Overview of Generative AI

This is a dummy section.

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